

III: Depth Estimation & Beyond

3D CV

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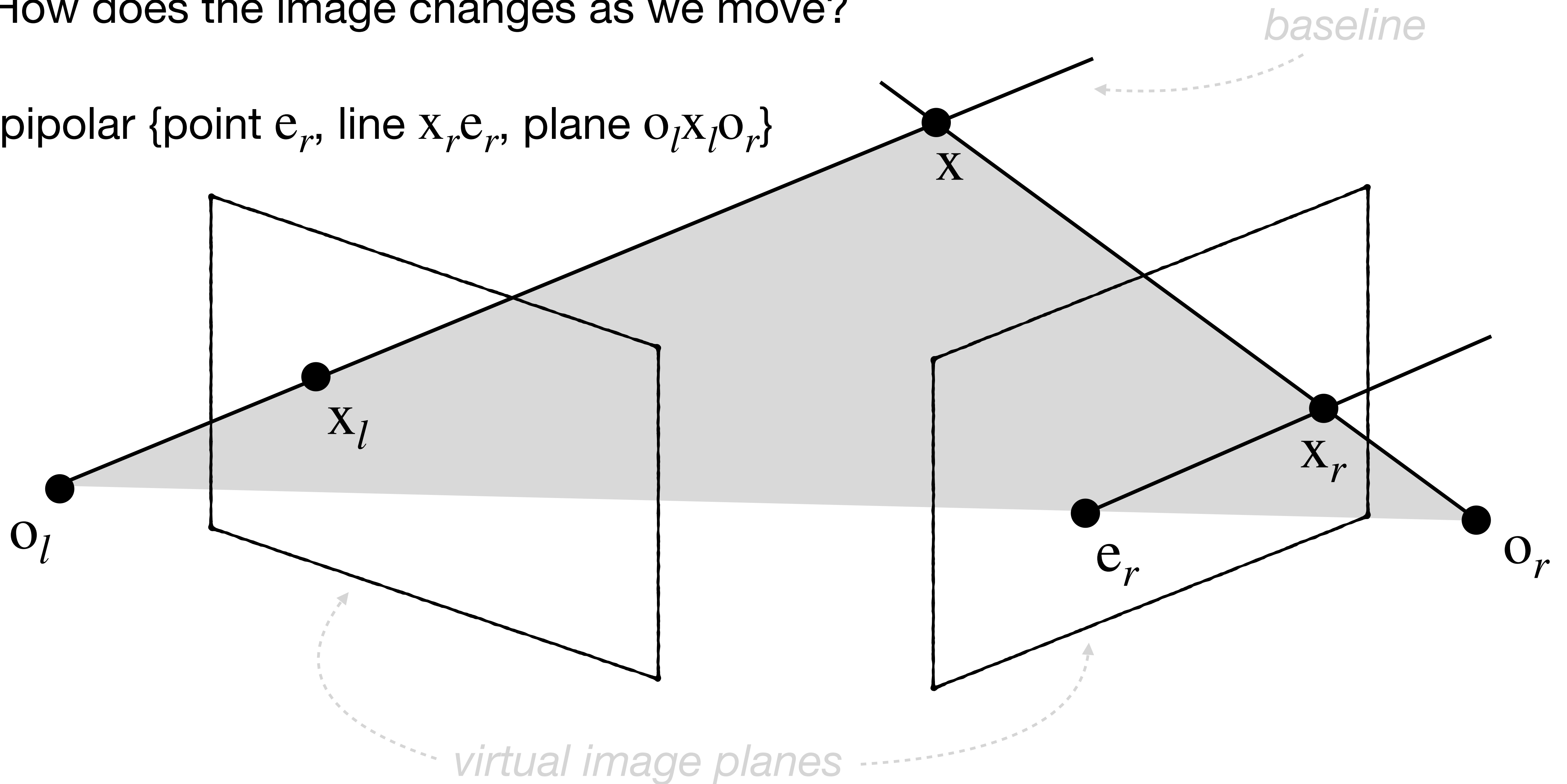
In the Previous Episode

- Two-view geometry
 - Epipolar geometry
 - Fundamental matrix
 - Triangulation
- Structure from motion

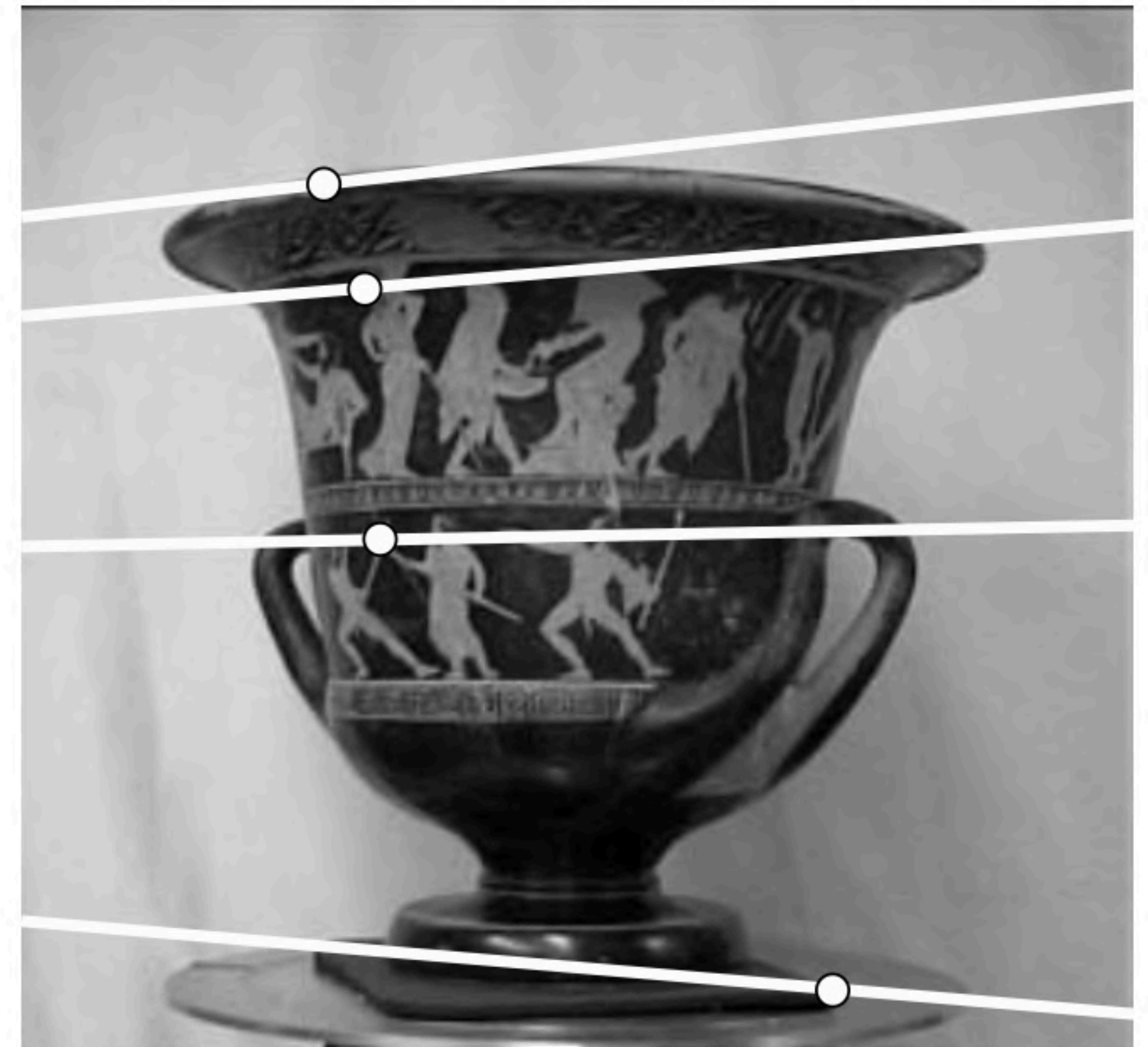
Epipolar Geometry

How does the image changes as we move?

epipolar {point e_r , line $x_r e_r$, plane $o_l x_l o_r$ }



Epipolar Lines in Real Life



Estimating Camera Positions

Using fundamental matrix F

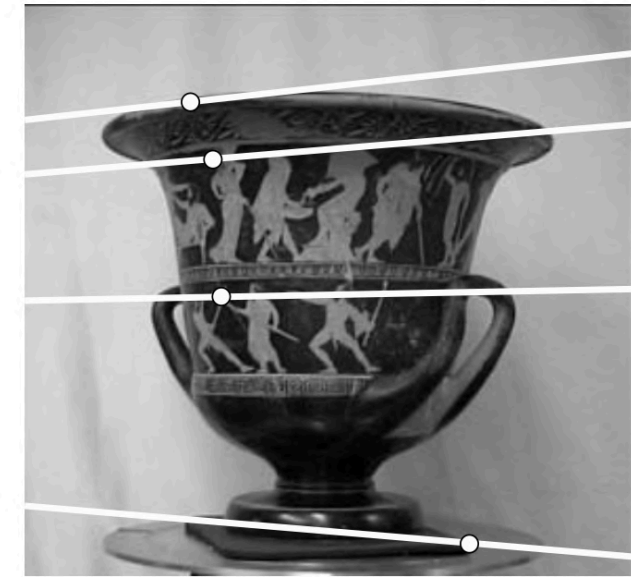
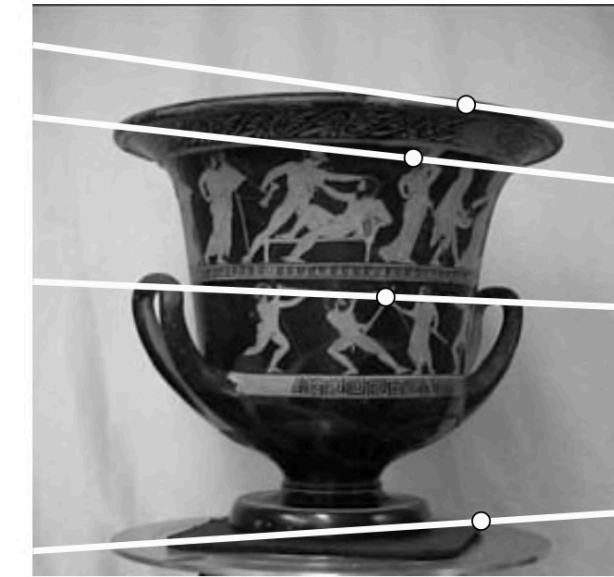
- Estimate F using correspondences

- Consider $H_l = [K_l | 0] \begin{bmatrix} I & 0 \\ 0 & 1 \end{bmatrix}$ and $H_r = [K_r | 0] \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix}$

- Fundamental matrix encodes R and t:

$$F = K_r^{-T} [t]_{\times} R K_l^{-1}$$

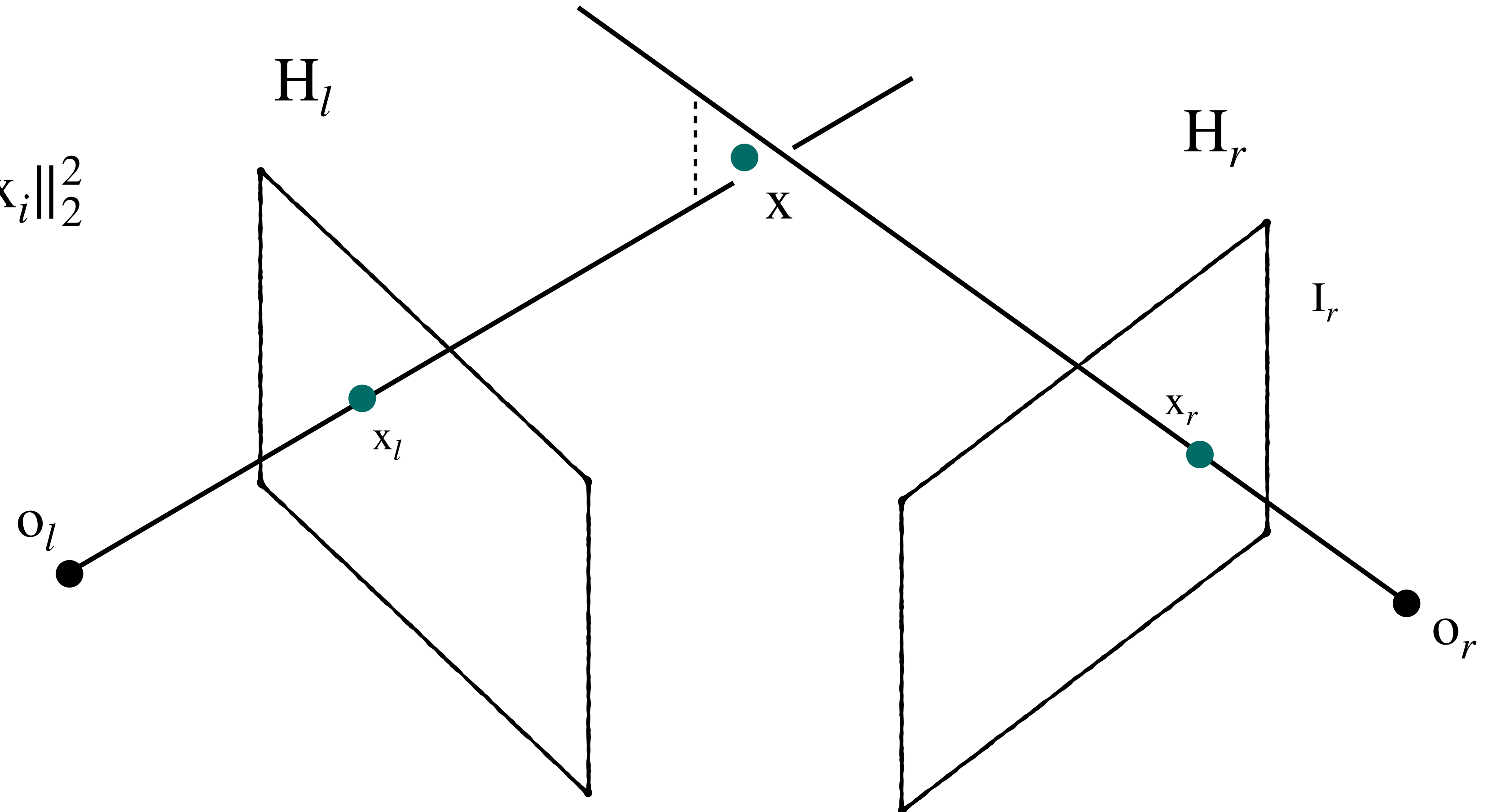
- Assuming K_r and K_l are known we can recover R and t



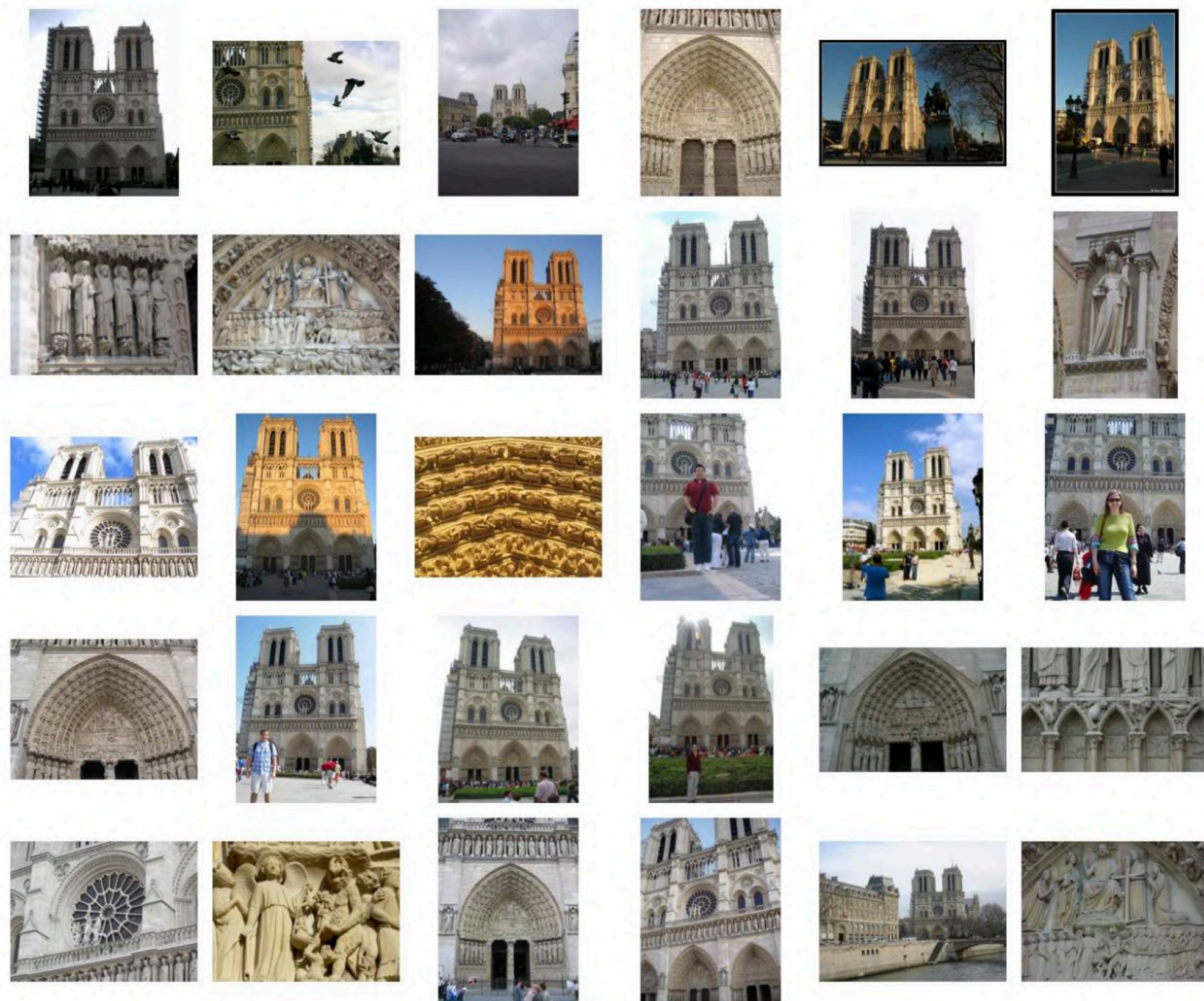
Structure Recovery

- Solve:

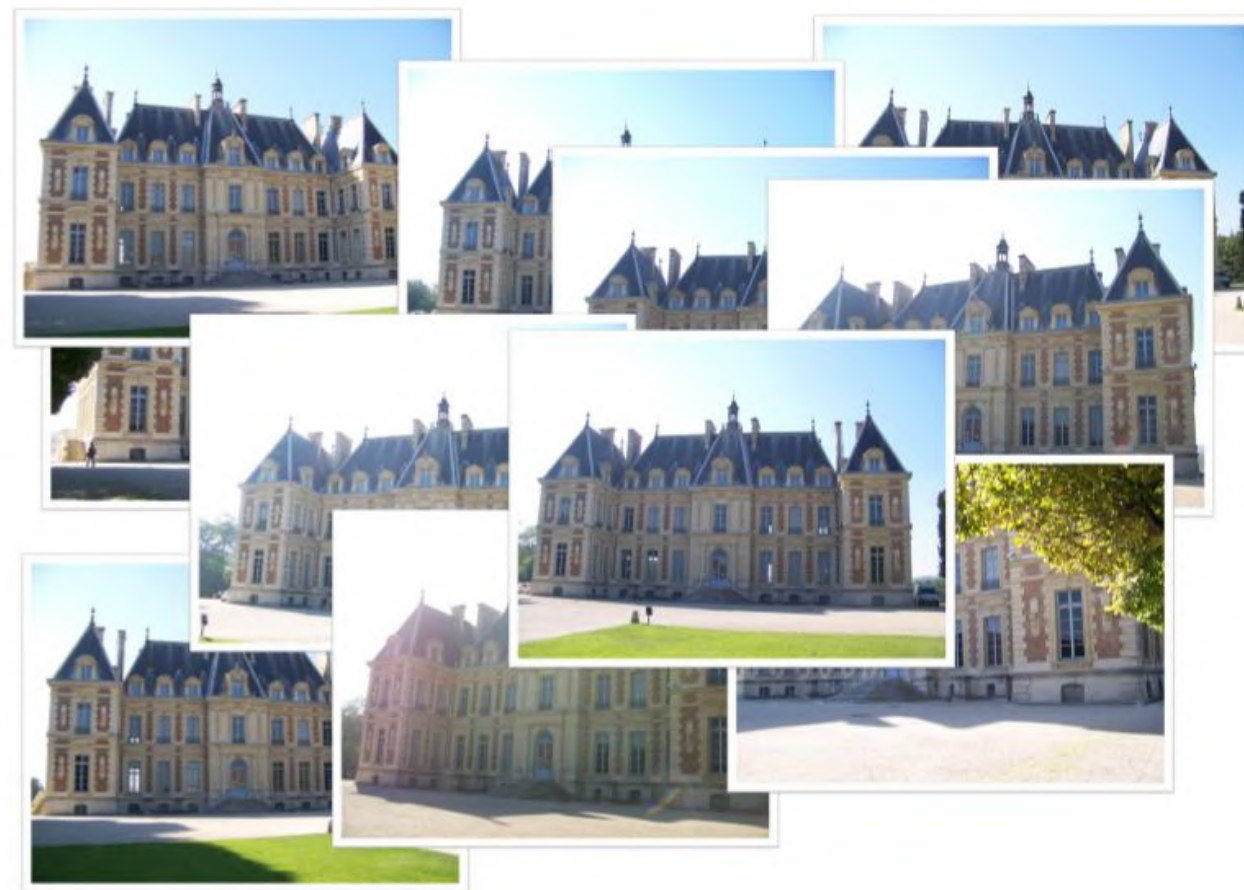
$$\mathbf{x} = \arg \min \sum_{i \in \{l, r\}} \|\mathbf{H}_i \mathbf{x} - \mathbf{x}_i\|_2^2$$



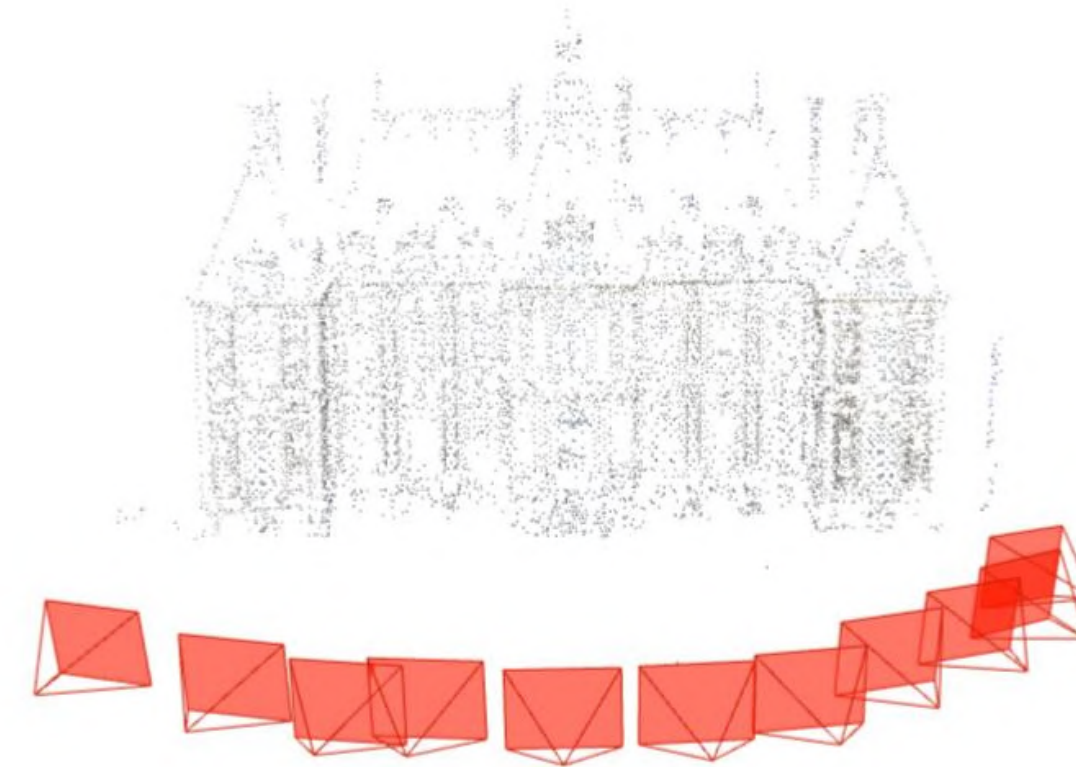
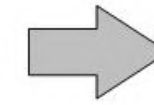
Structure from Motion



Today's Lecture: Depth Estimation



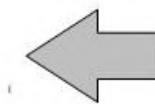
multi-view images



structure-from-motion



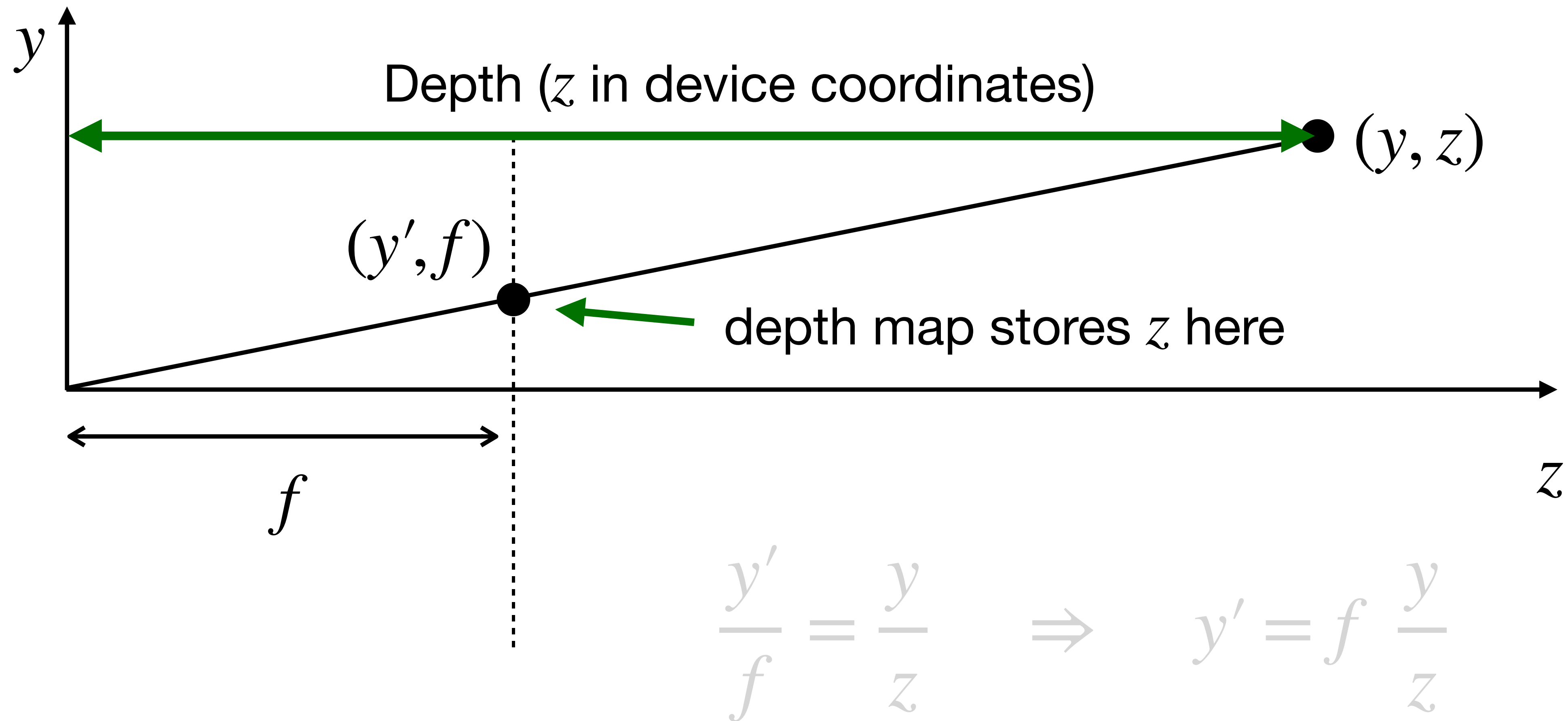
multi-view stereo



surface reconstruction

Depth Estimation Basics

Depth and Depth Maps



From 2D to 2.5D

- Consider $\mathbf{x} = (x, y, z, 1)$
- Project

$$[\mathbf{K} \mid 0] \mathbf{x} = \begin{pmatrix} f_x & 0 & p_x & 0 \\ 0 & f_y & p_y & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \mathbf{x} = \begin{pmatrix} f_x x + p_x z \\ f_y y + p_y z \\ z \end{pmatrix}$$

- Find inhomogeneous coordinates

$$\begin{pmatrix} f_x x + p_x z \\ f_y y + p_y z \\ z \end{pmatrix} \rightarrow \begin{pmatrix} x_{pix} \\ y_{pix} \\ 1 \end{pmatrix}$$

3D



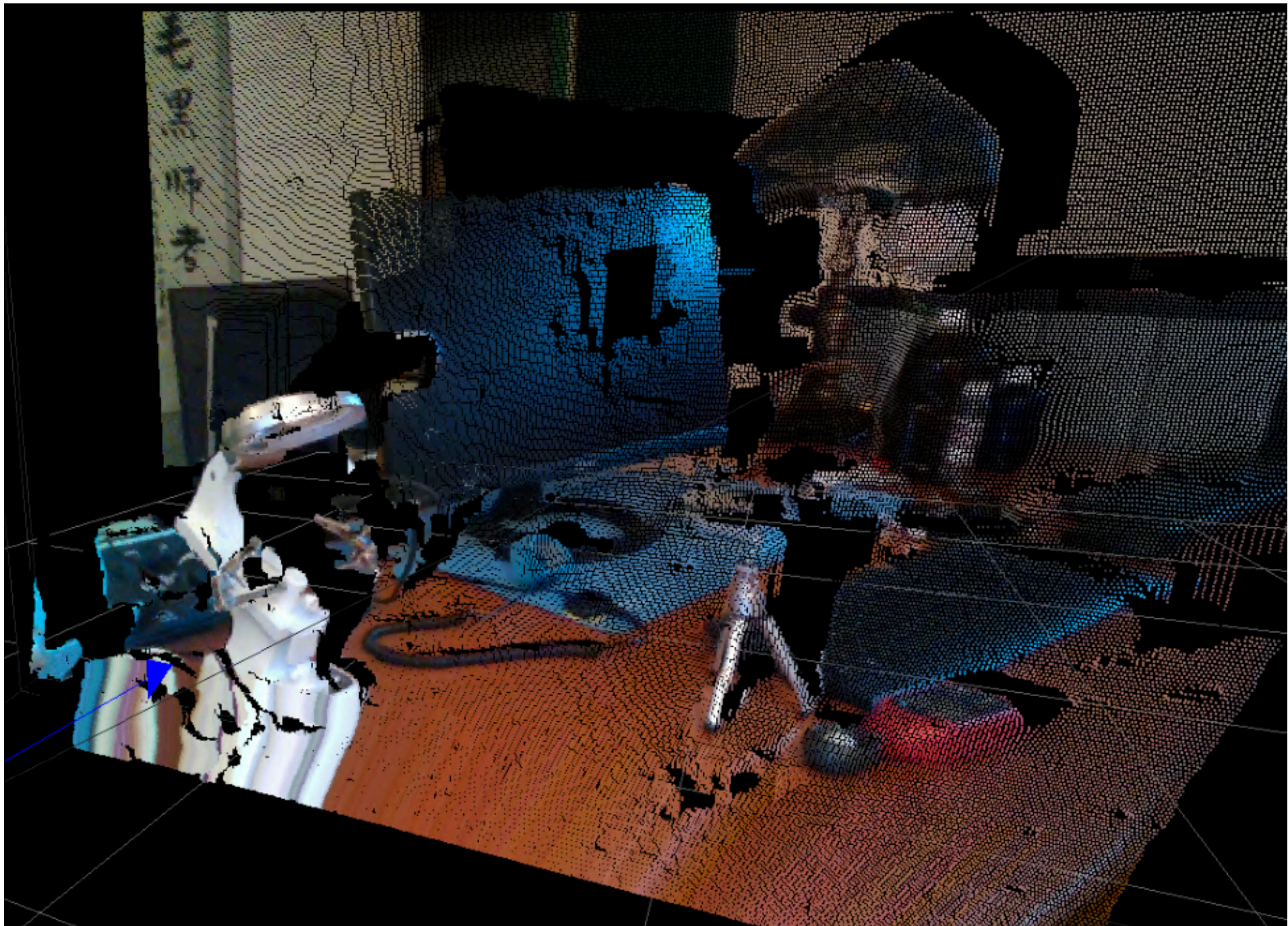
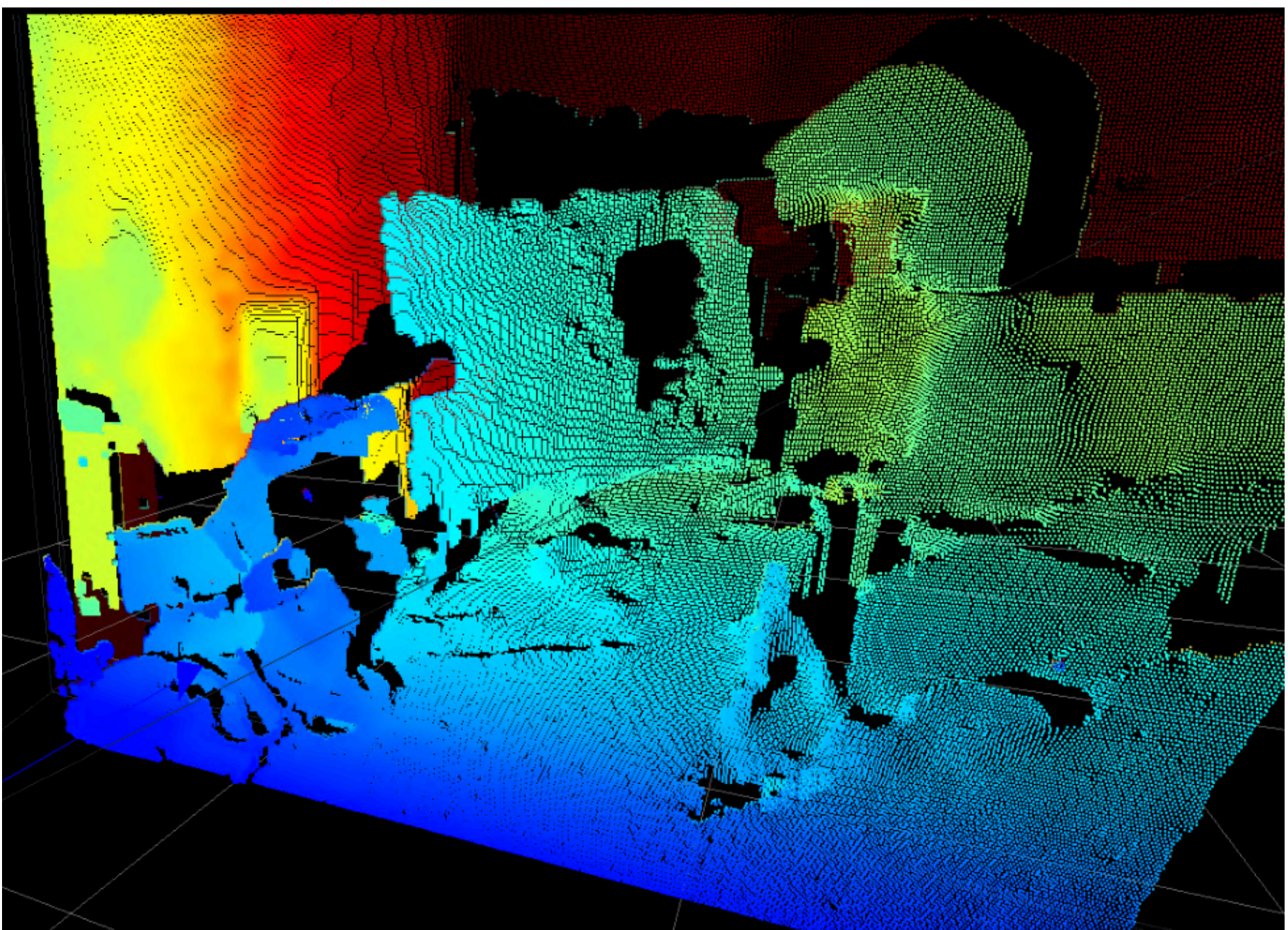
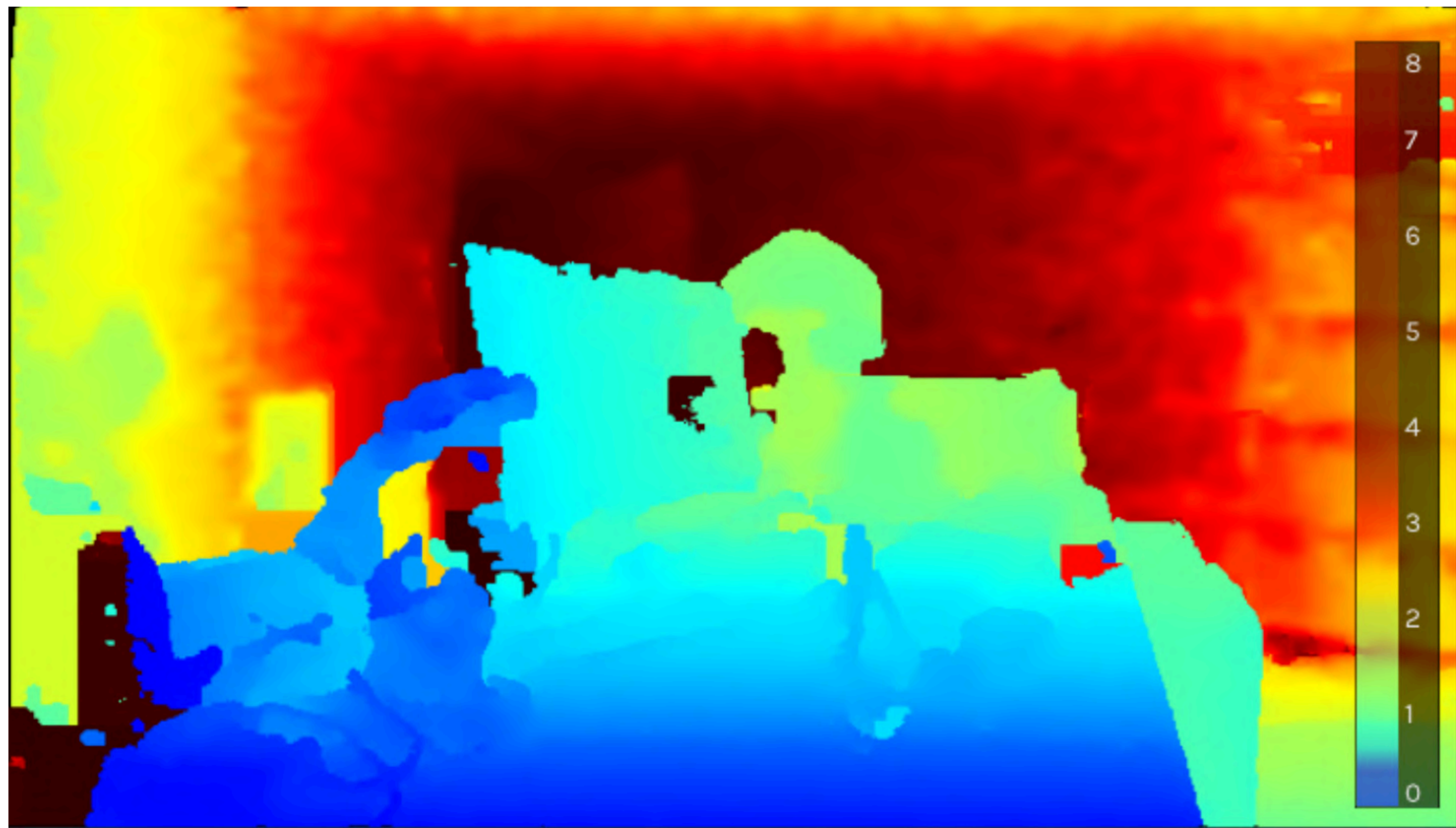
2D

- Back projected point is $z \mathbf{K}^{-1} \begin{pmatrix} x_{pix} \\ y_{pix} \\ 1 \end{pmatrix}$
- Reverse projection

$$\mathbf{K}^{-1} \begin{pmatrix} z \cdot x_{pix} \\ z \cdot y_{pix} \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

- Get homogeneous coordinates

$$\begin{pmatrix} x_{pix} \\ y_{pix} \end{pmatrix} \rightarrow z \begin{pmatrix} x_{pix} \\ y_{pix} \\ 1 \end{pmatrix}$$



Binocular Vision

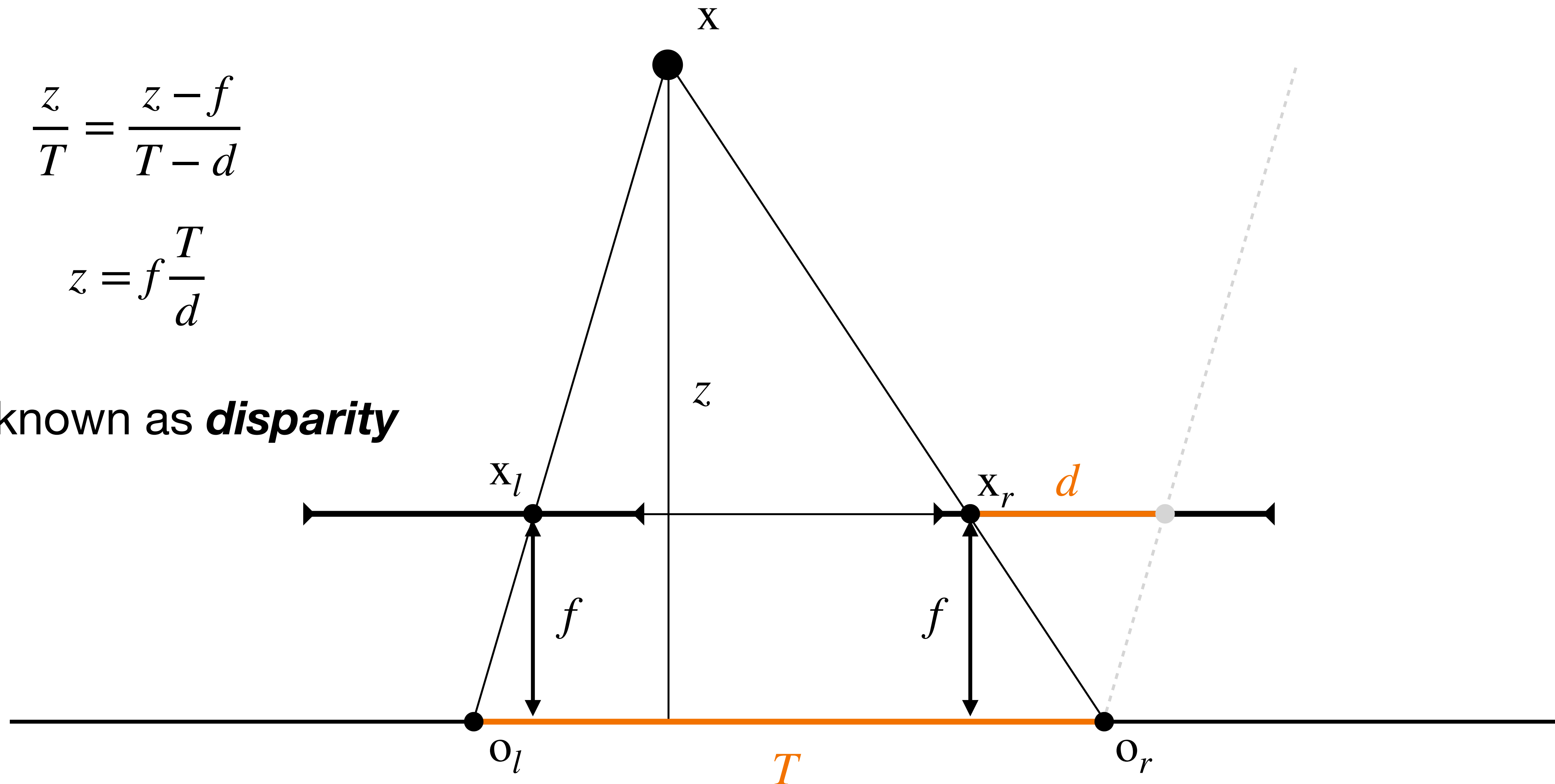


Depth Estimation in Two-View Geometry

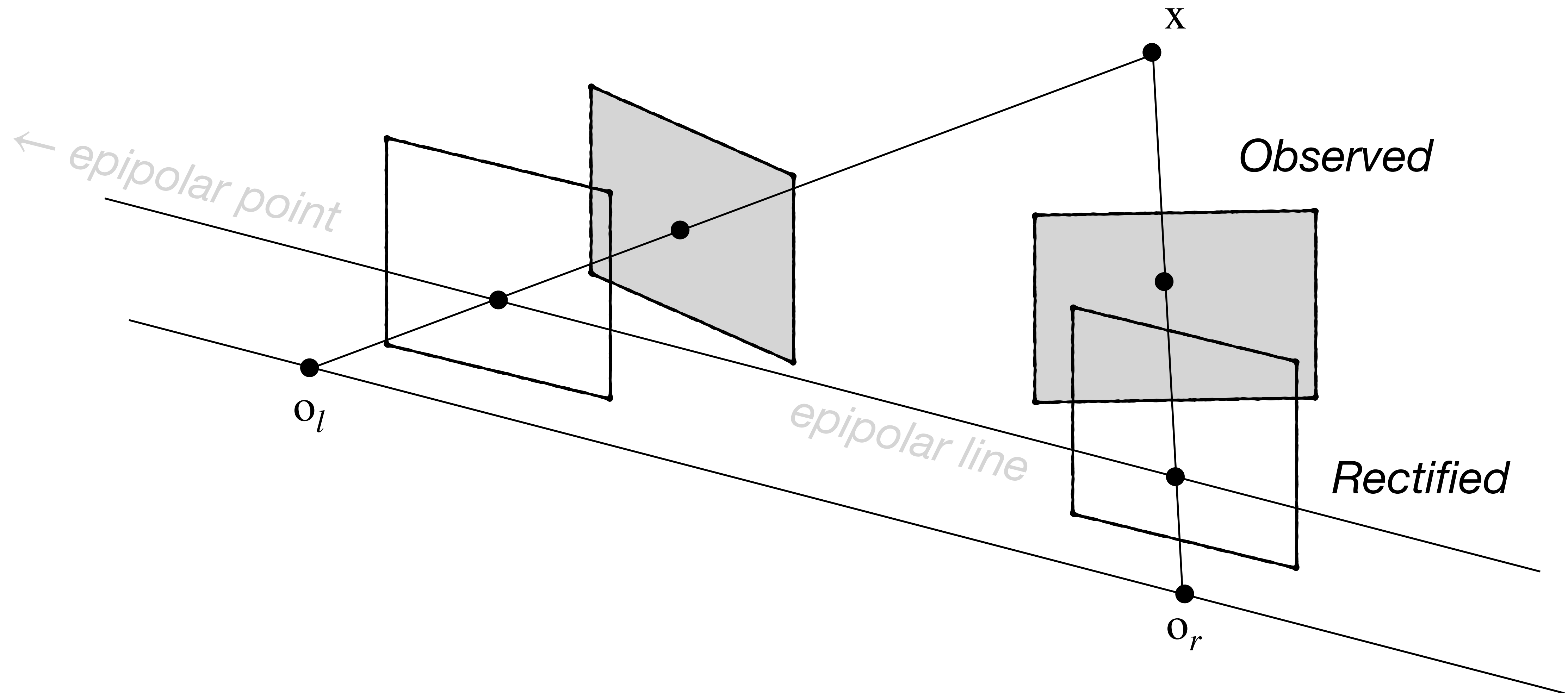
$$\frac{z}{T} = \frac{z - f}{T - d}$$

$$z = f \frac{T}{d}$$

d is known as **disparity**



Rectification



Rectification Example



Difficulties in Estimating Disparity



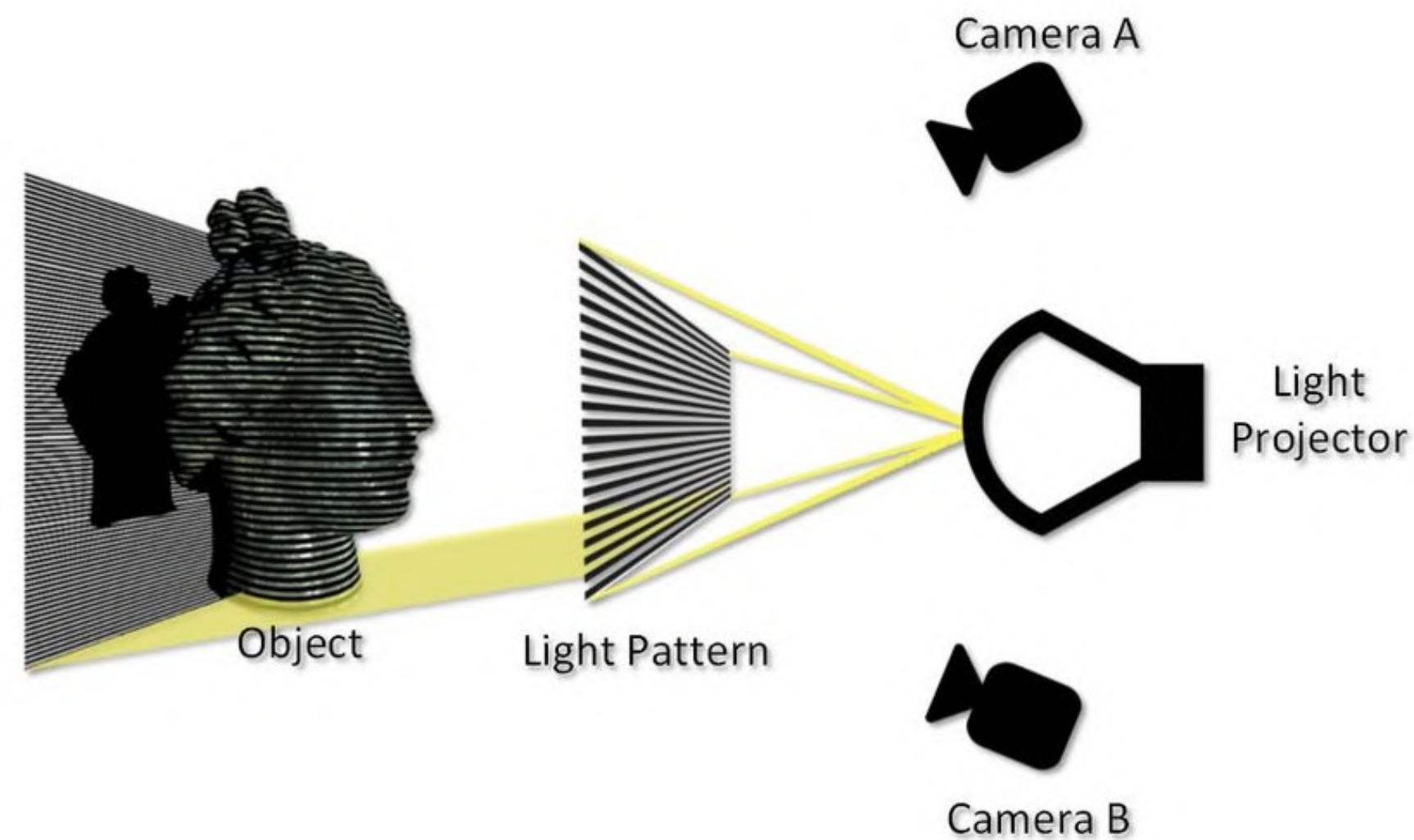
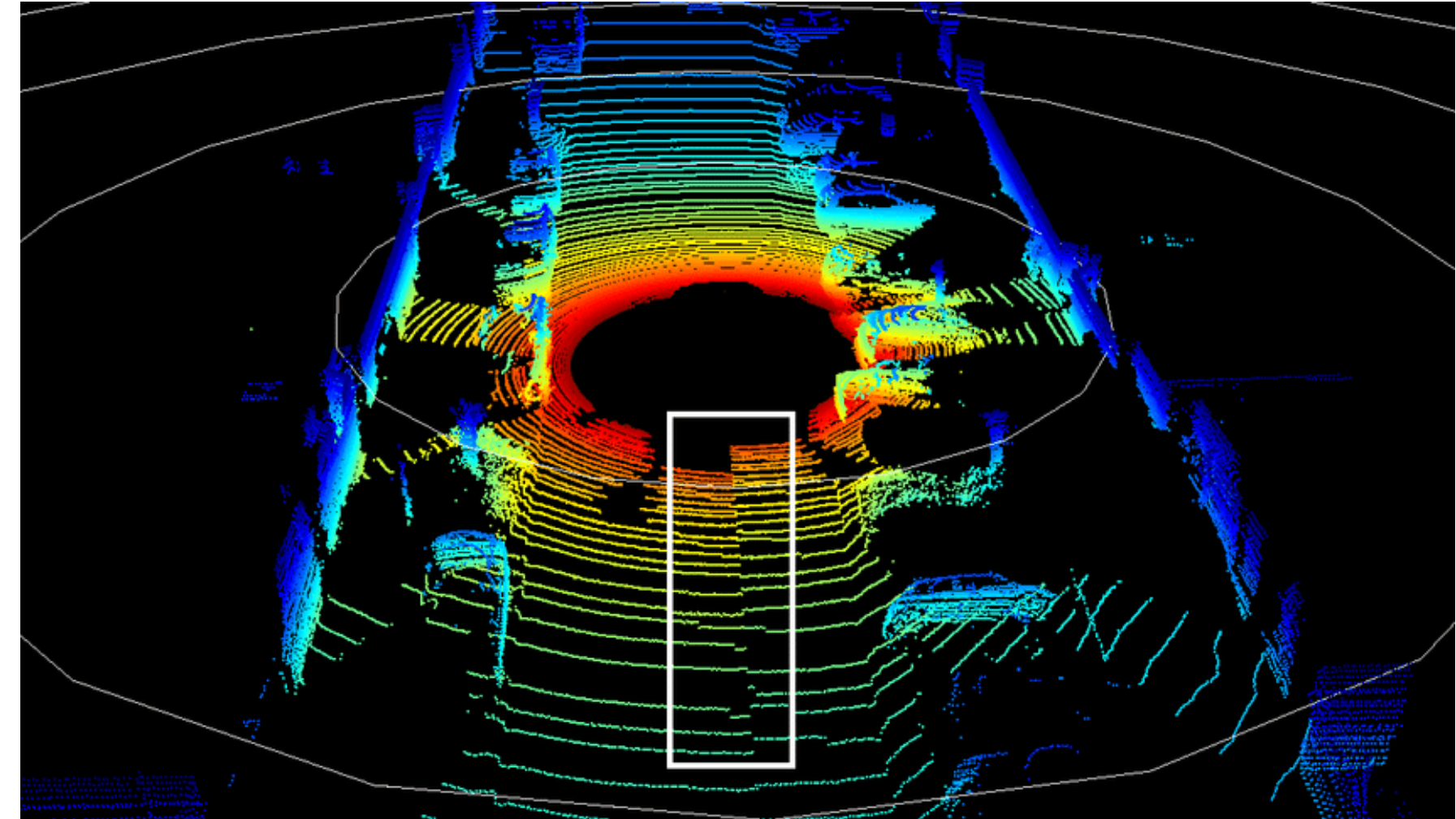
Learning-Based Approaches

- Heuristics for disparity estimation have limited success
- We perceive depth even without binocular vision
 - There are many cues around us



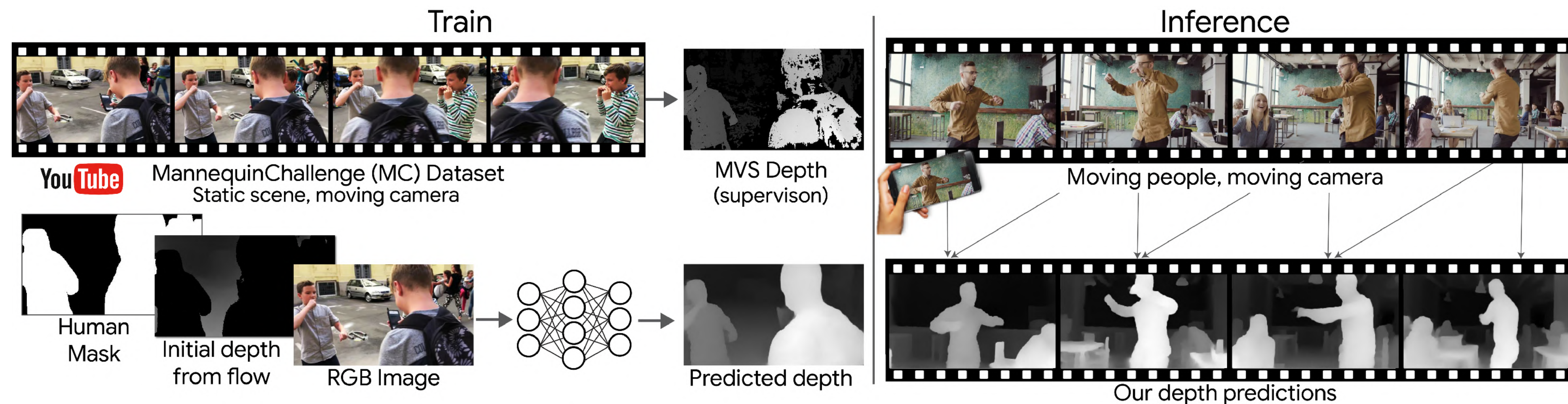
Data

- Input: mono / stereo images
- Target: depth sensors
 - Structured light
 - Time of flight
 - Lidar
- Target: SfM
- Target: synthetic



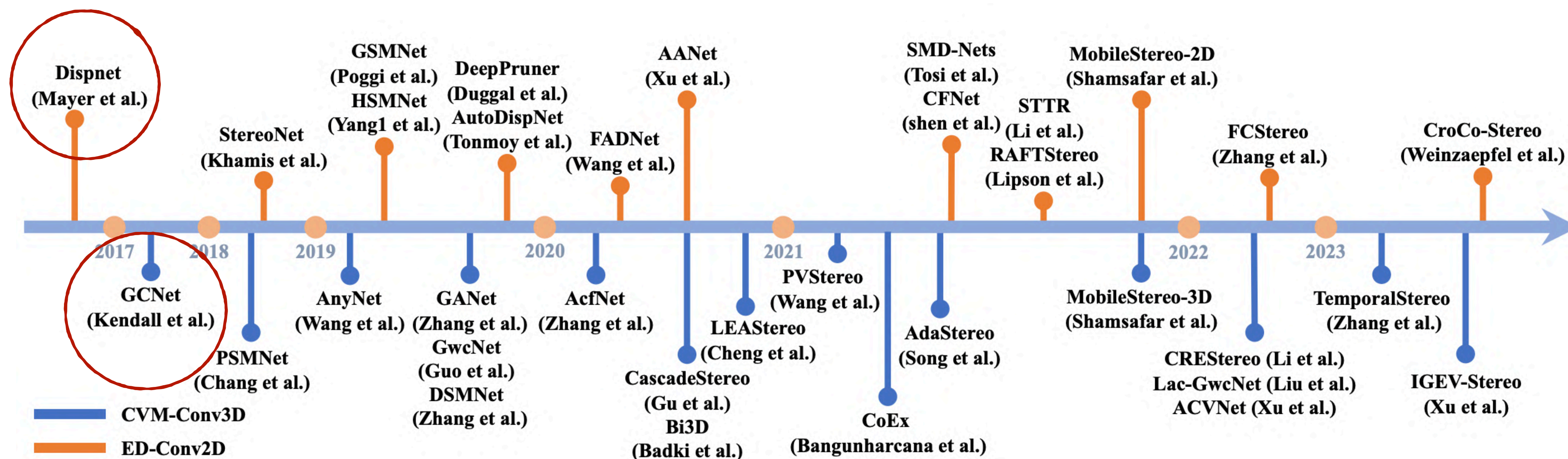
2026 is the new 2016

- Dense depth of moving people are hard to collect
- Mannequin challenge: people are frozen while camera tours the scene
- Idea (2019): run SfM on videos to get data



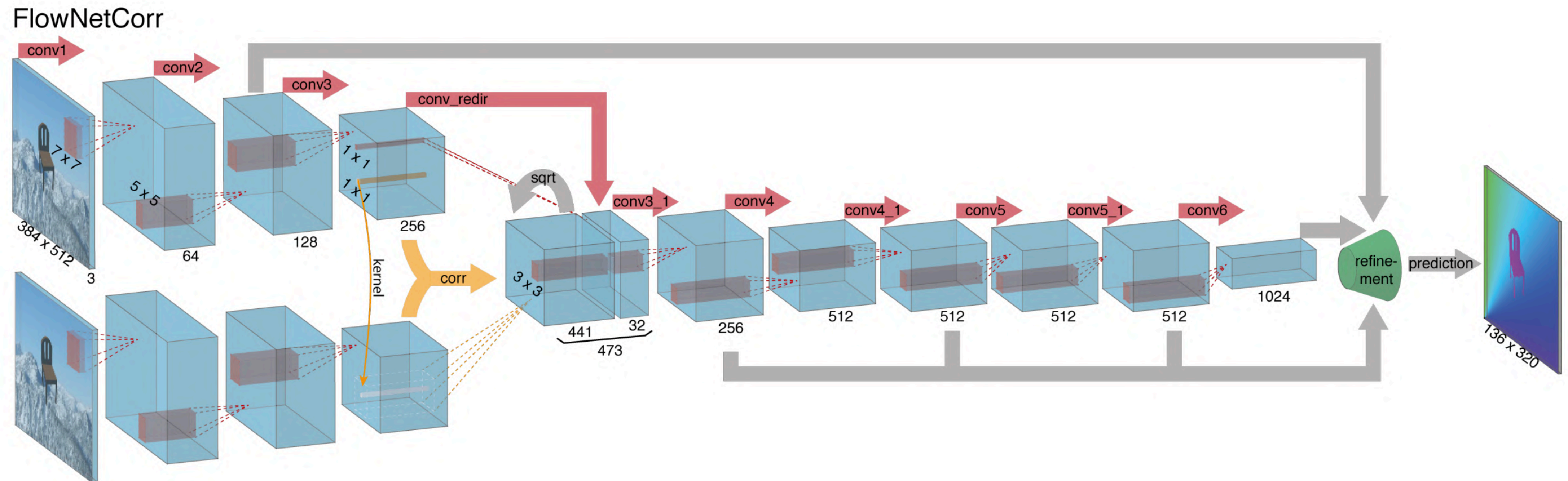
Stereo Depth: Disparity Estimation

fast, based on 2D convolutions



high precision, uses 3D convolutions

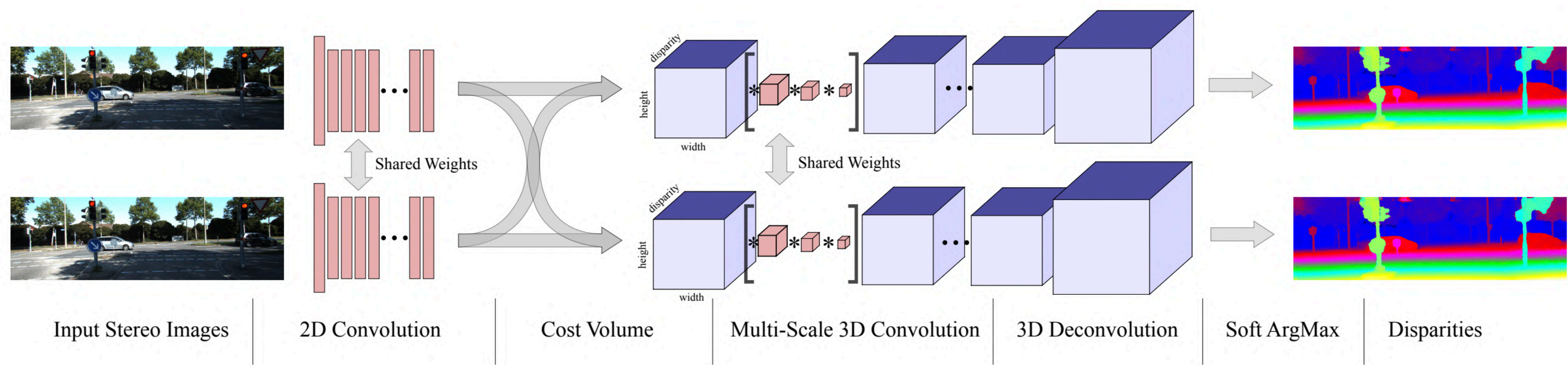
DispNet



Dosovitskiy A. et al. FlowNet: Learning optical flow with convolutional networks

Mayer N. et al. A large dataset to train convolutional networks for disparity, optical flow, and scene flow estimation

GCNet



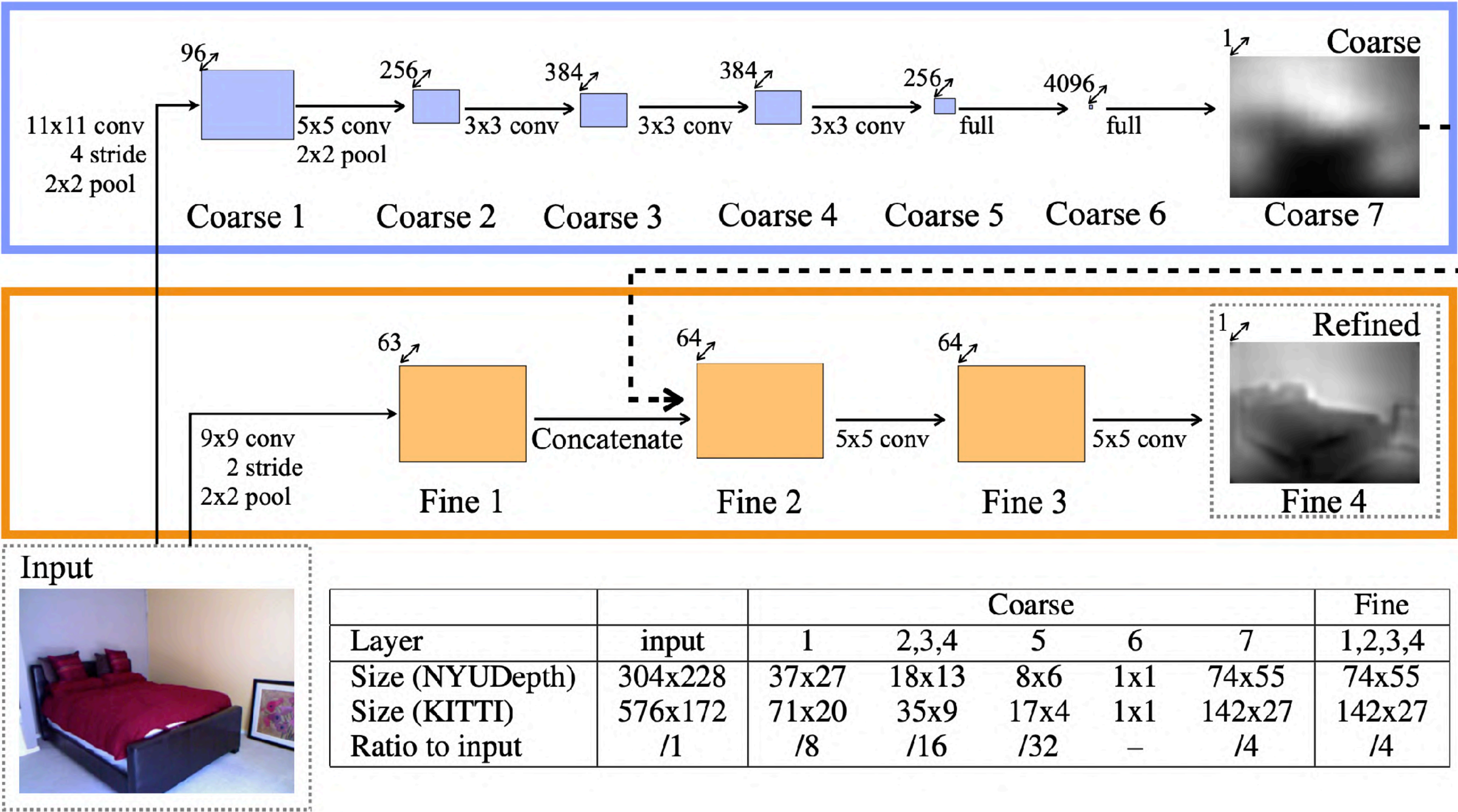
Monocular Depth Estimation

Depth Map Prediction

from a Single Image using a Multi-Scale Deep Network

- Scale invariant error for depths y and y^*
- Denote $d_i = \log y_i - \log y_i^*$

$$\begin{aligned}
 D(y, y^*) &= \frac{1}{n^2} \sum_{i,j} \left((\log y_i - \log y_j) - (\log y_i^* - \log y_j^*) \right)^2 \\
 &= \frac{1}{n} \sum_i d_i^2 - \frac{1}{n^2} \sum_{i,j} d_i d_j \\
 &= \frac{1}{n} \sum_i d_i^2 - \frac{\lambda}{n^2} \left(\sum_i d_i \right)^2
 \end{aligned}$$



Depth Map Prediction

from a Single Image using a Multi-Scale Deep Network



Towards Robust Monocular Depth Estimation

Mixing Datasets for Zero-Shot Cross-Dataset Transfer

Dataset	Indoor	Outdoor	Dynamic	Video	Dense	Accuracy	Diversity	Annotation	Depth	# Images
DIML Indoor [31]	✓			✓	✓	Medium	Medium	RGB-D	Metric	220K
MegaDepth [11]		✓	(✓)		(✓)	Medium	Medium	SfM	No scale	130K
ReDWeb [32]	✓	✓	✓		✓	Medium	High	Stereo	No scale & shift	3600
WSVD [33]	✓	✓	✓	✓	✓	Medium	High	Stereo	No scale & shift	1.5M
3D Movies	✓	✓	✓	✓	✓	Medium	High	Stereo	No scale & shift	75K
DIW [34]	✓	✓	✓			Low	High	User clicks	Ordinal pair	496K
ETH3D [35]	✓	✓			✓	High	Low	Laser	Metric	454
Sintel [36]	✓	✓	✓	✓	✓	High	Medium	Synthetic	(Metric)	1064
KITTI [28], [29]		✓	(✓)	✓	(✓)	Medium	Low	Laser/Stereo	Metric	93K
NYUDv2 [30]	✓		(✓)	✓	✓	Medium	Low	RGB-D	Metric	407K
TUM-RGBD [37]	✓		(✓)	✓	✓	Medium	Low	RGB-D	Metric	80K

Towards Robust Monocular Depth Estimation

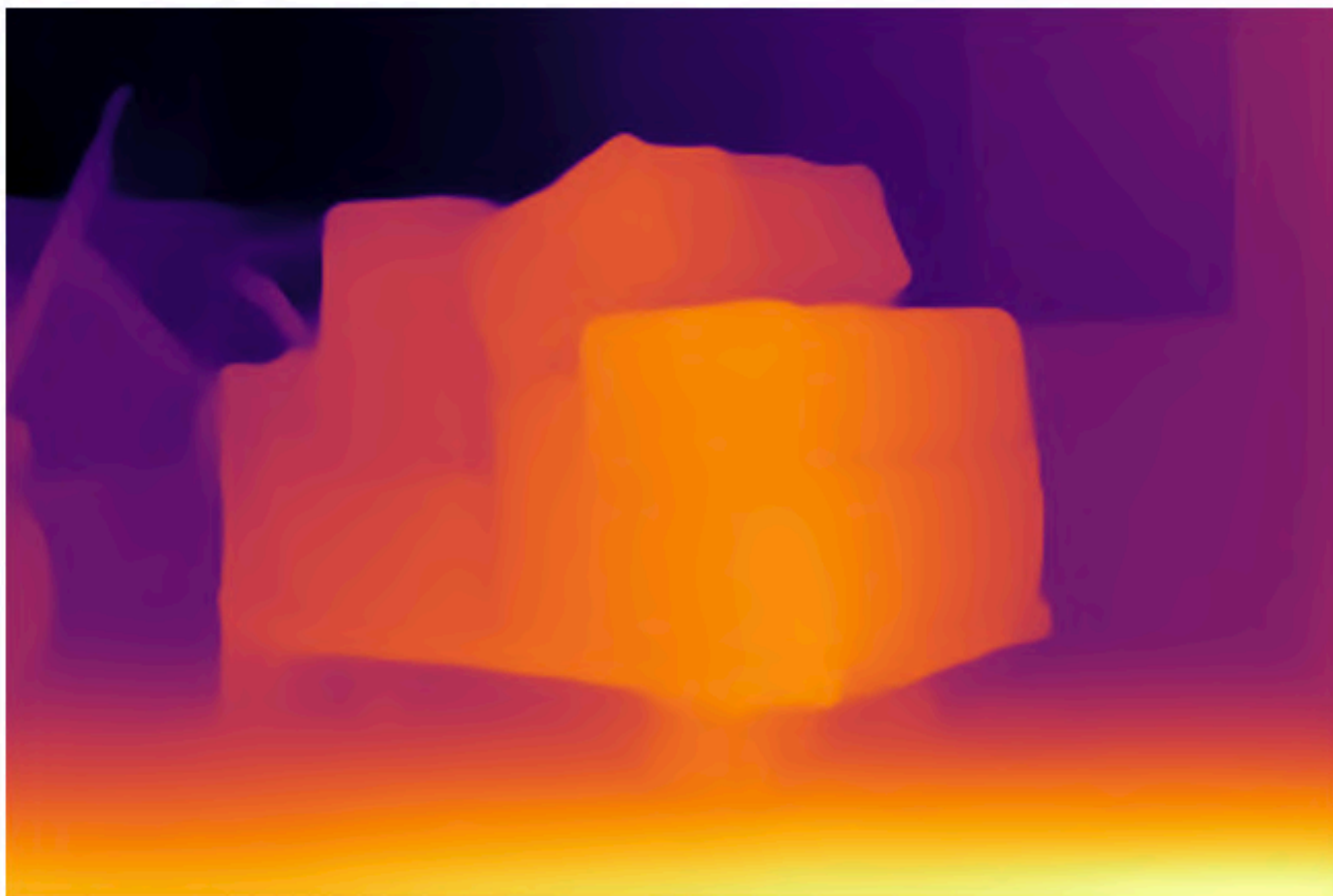
Mixing Datasets for Zero-Shot Cross-Dataset Transfer

- In some datasets, intrinsic parameters of the camera are unknown
- As a result, disparity (target) is *shifted*
- MiDaS: predict disparity *up to scale and shift*

$$L_{ssi}(\hat{d}, \hat{d}^*) = \frac{1}{2M} \sum_{i=1}^M \rho(\hat{d}_i - \hat{d}_i^*)$$

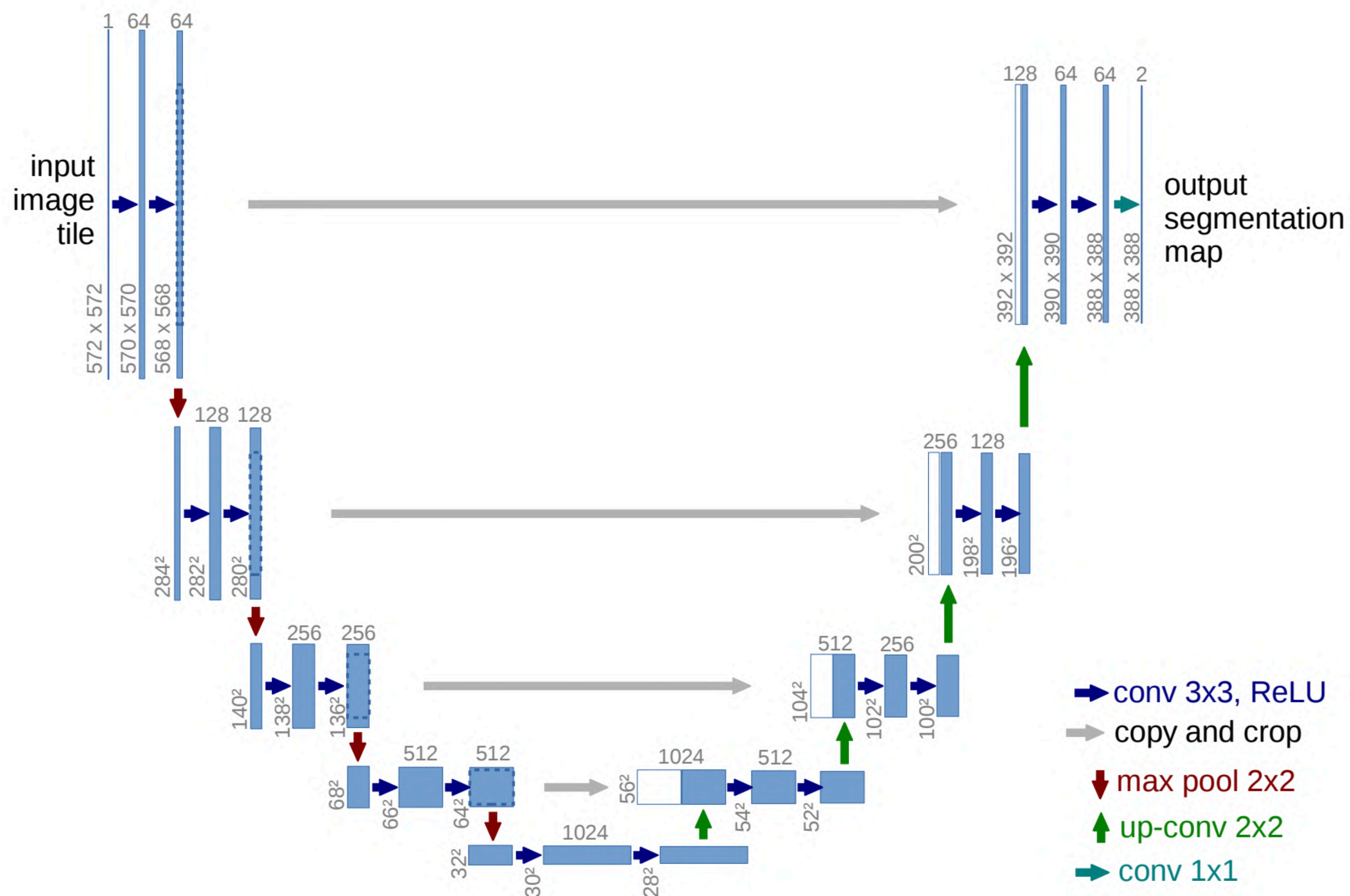
- Where $\hat{d}_i = sd_i + t$ and $\hat{d}_i^* = d_i^*$ for $(s, t) = \arg \min_{s', t'} \sum_i (s'd_i + t' - d_i^*)^2$

- Architecture similar to U-Net

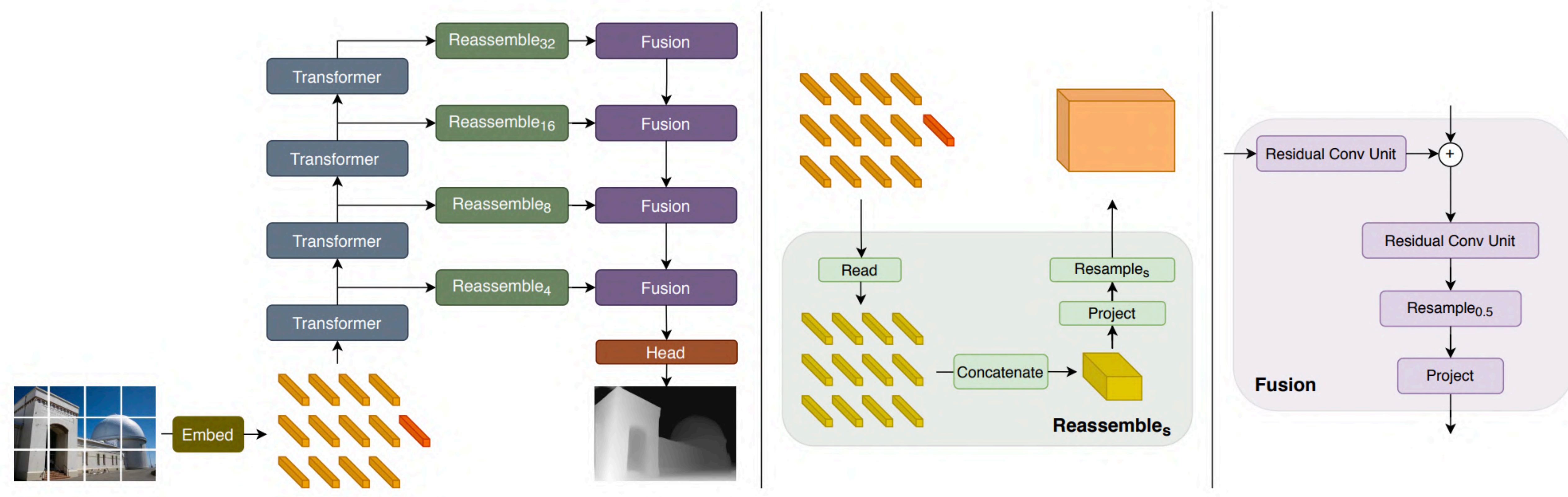


Dense Prediction Architectures

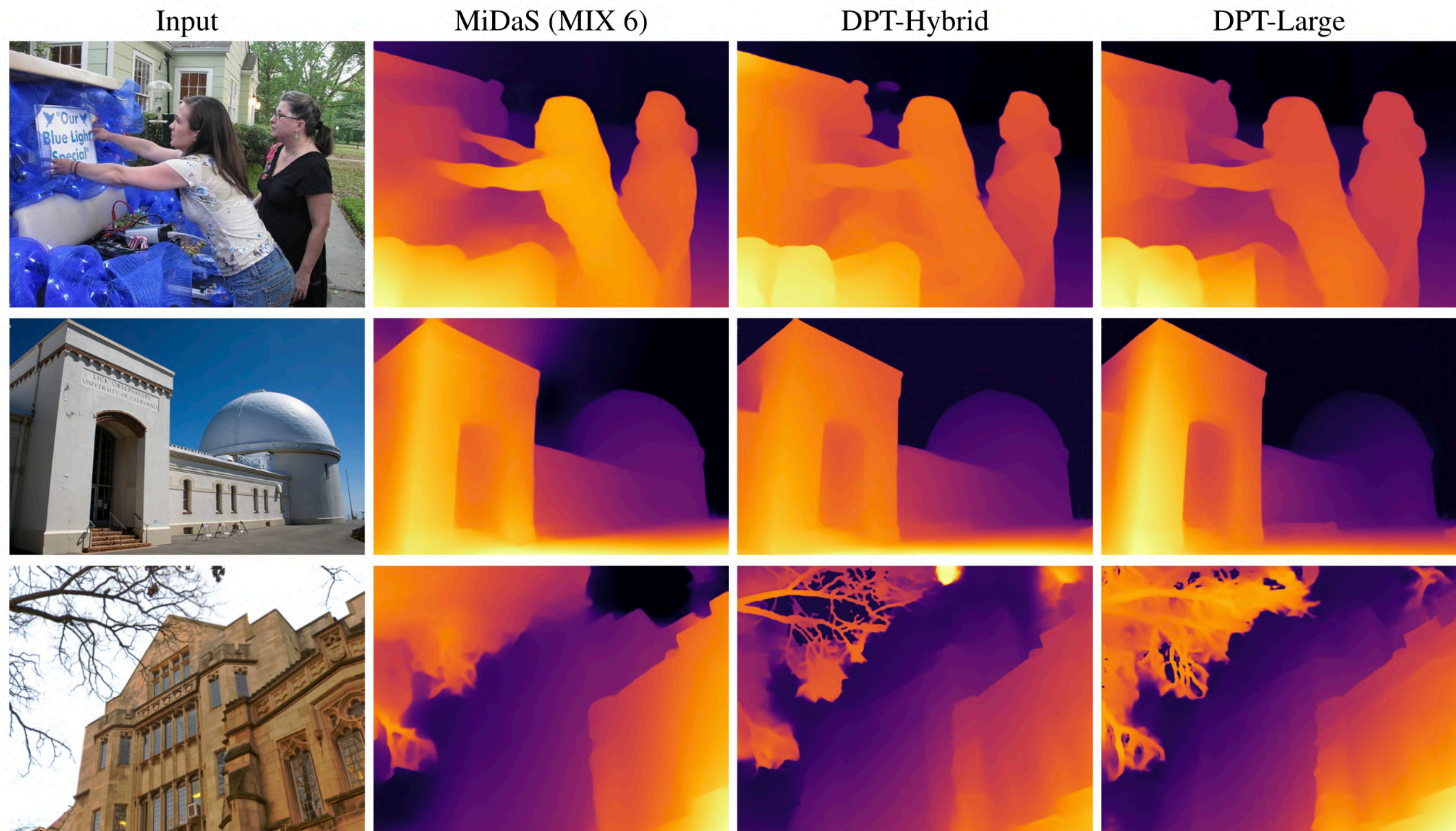
Originally MiDAS used UNets



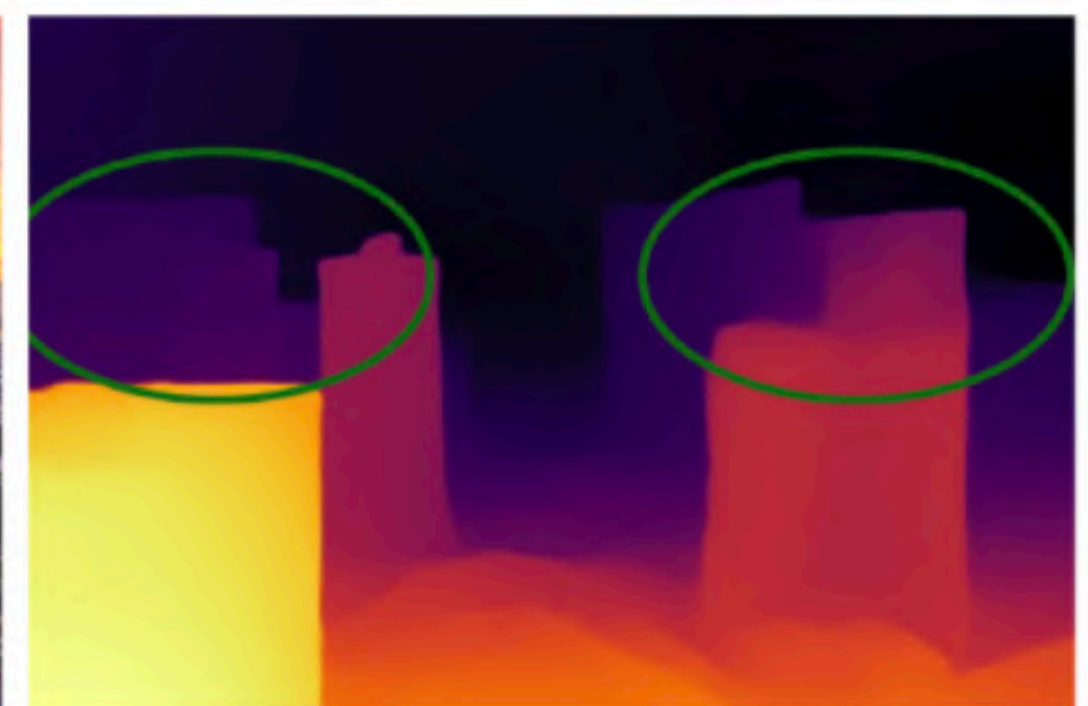
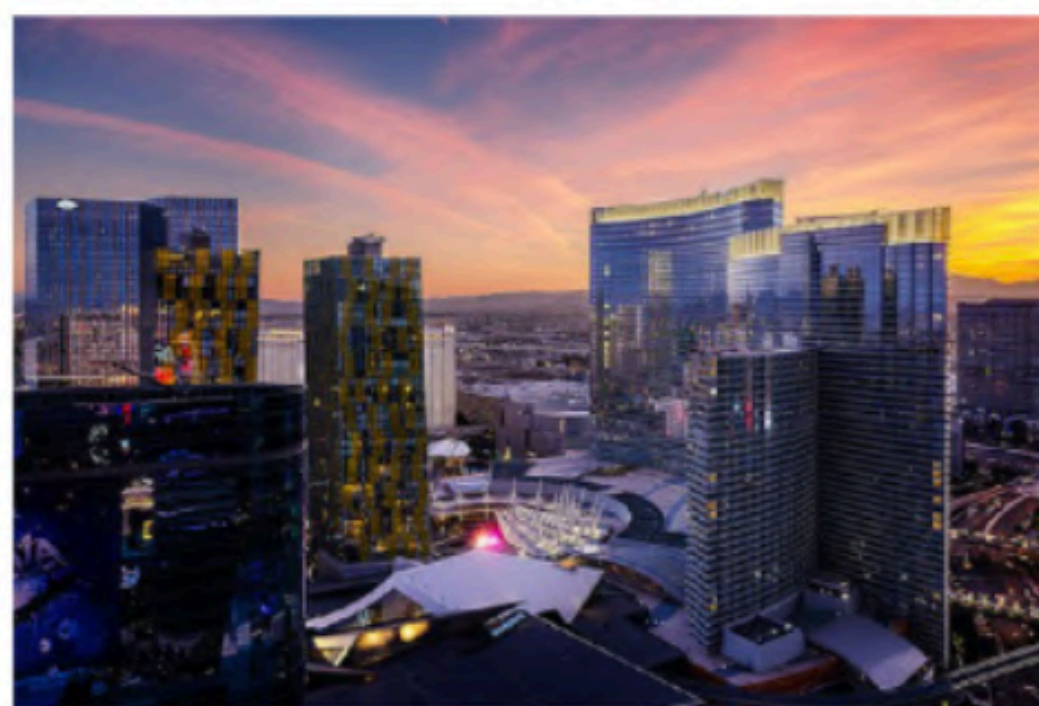
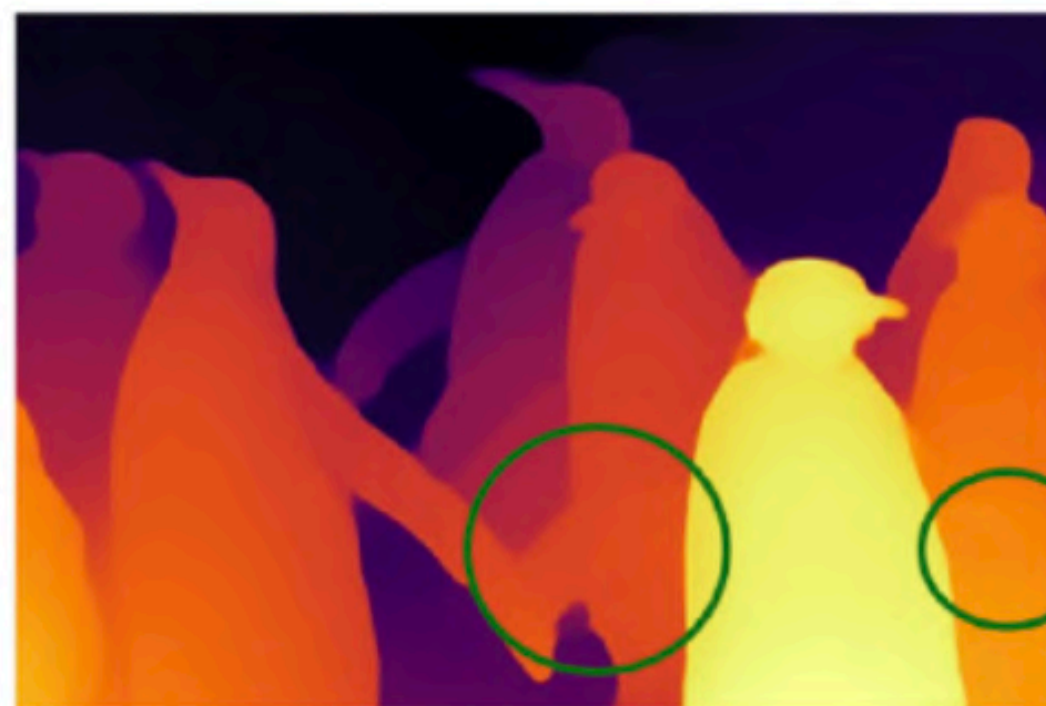
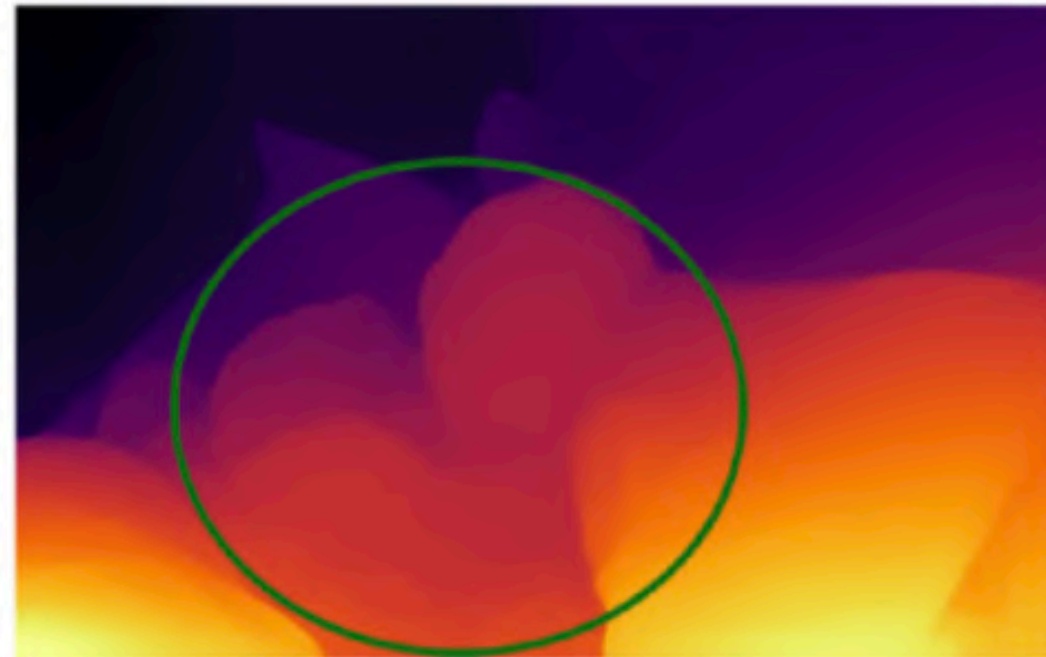
Vision Transformers for Dense Prediction



From Convolutions to Transformers

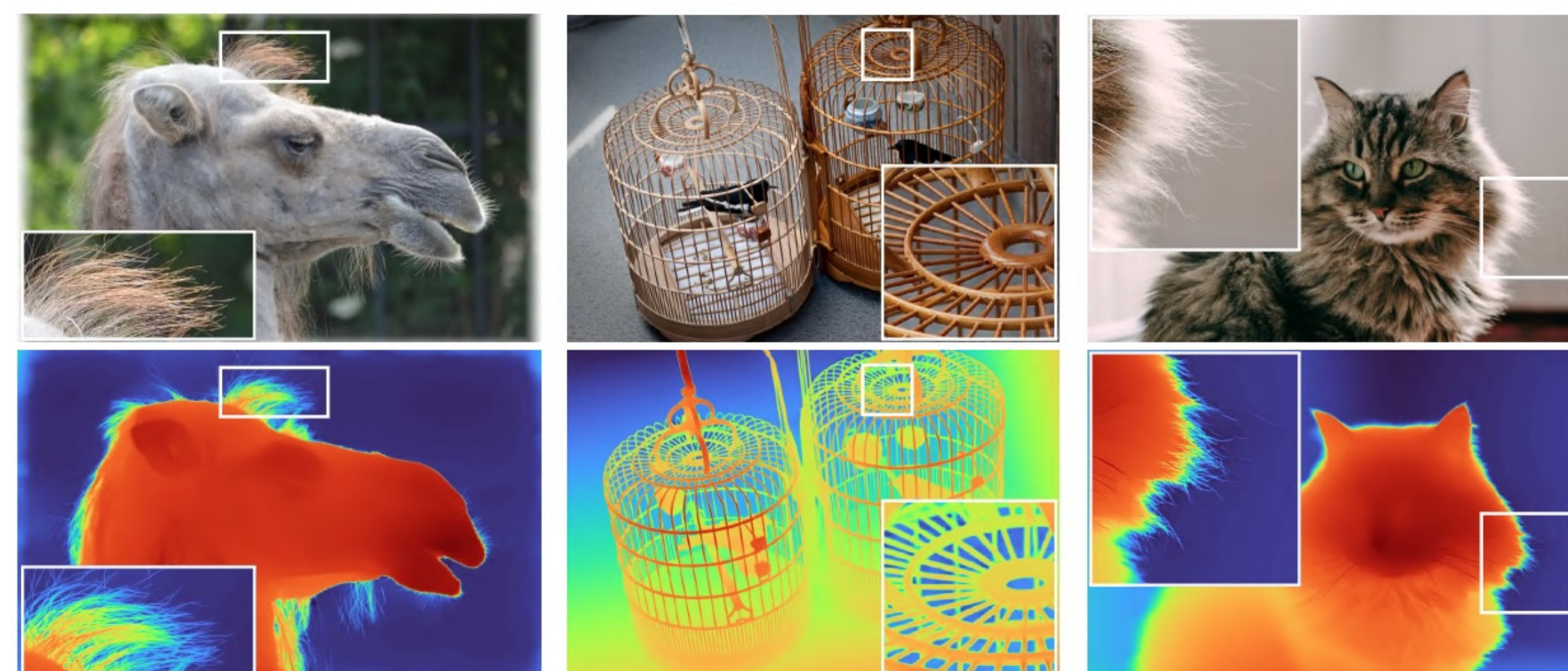


Failure Cases of MiDaS



Recent Notable Works

- **Marigold:** repurpose stable diffusion as a backbone from depth estimation
- **Depth Anything:** scale data by leveraging 62m unlabelled images
- **Depth-Pro:** multi-scale transformer architecture aiming for metric depth estimation with sharp border prediction



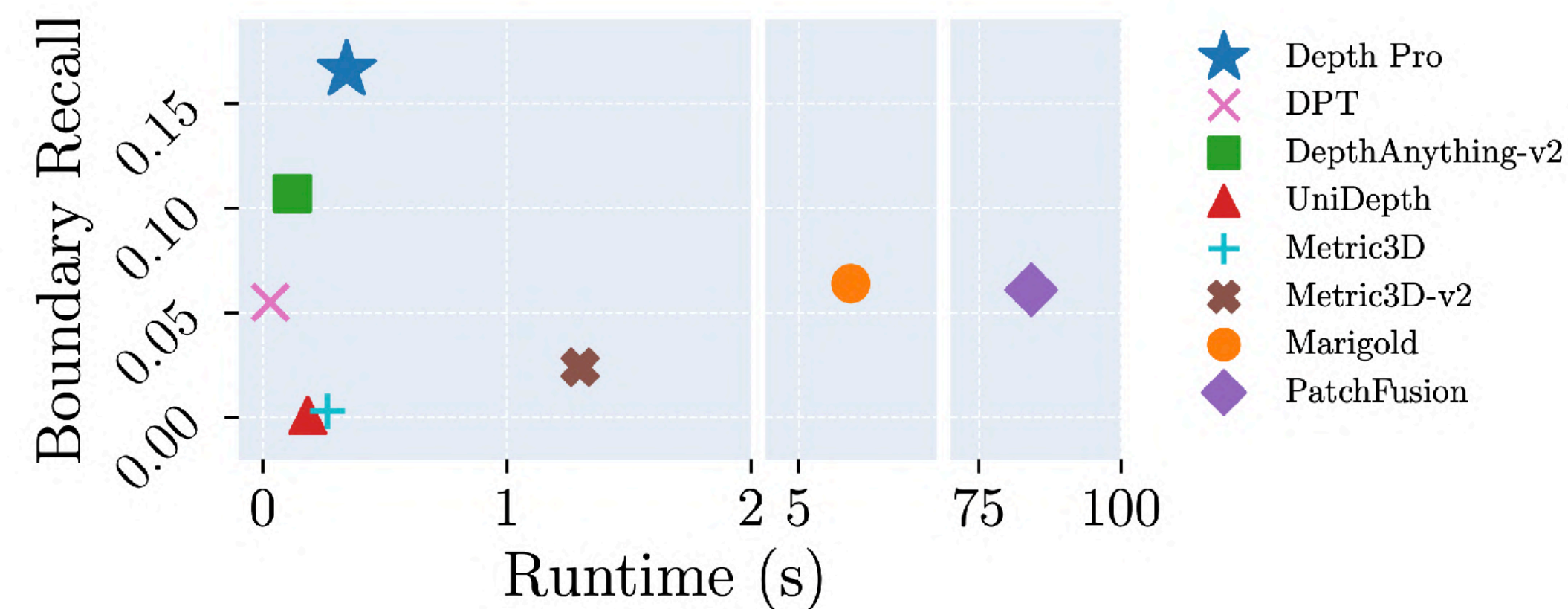
<https://github.com/prs-eth/Marigold>

<https://github.com/DepthAnything/Depth-Anything-V2>

<https://github.com/apple/ml-depth-pro>

Recent Progress

δ_1 metric reports percent of points for which the predicted and ground-truth depths are within 25% of each other

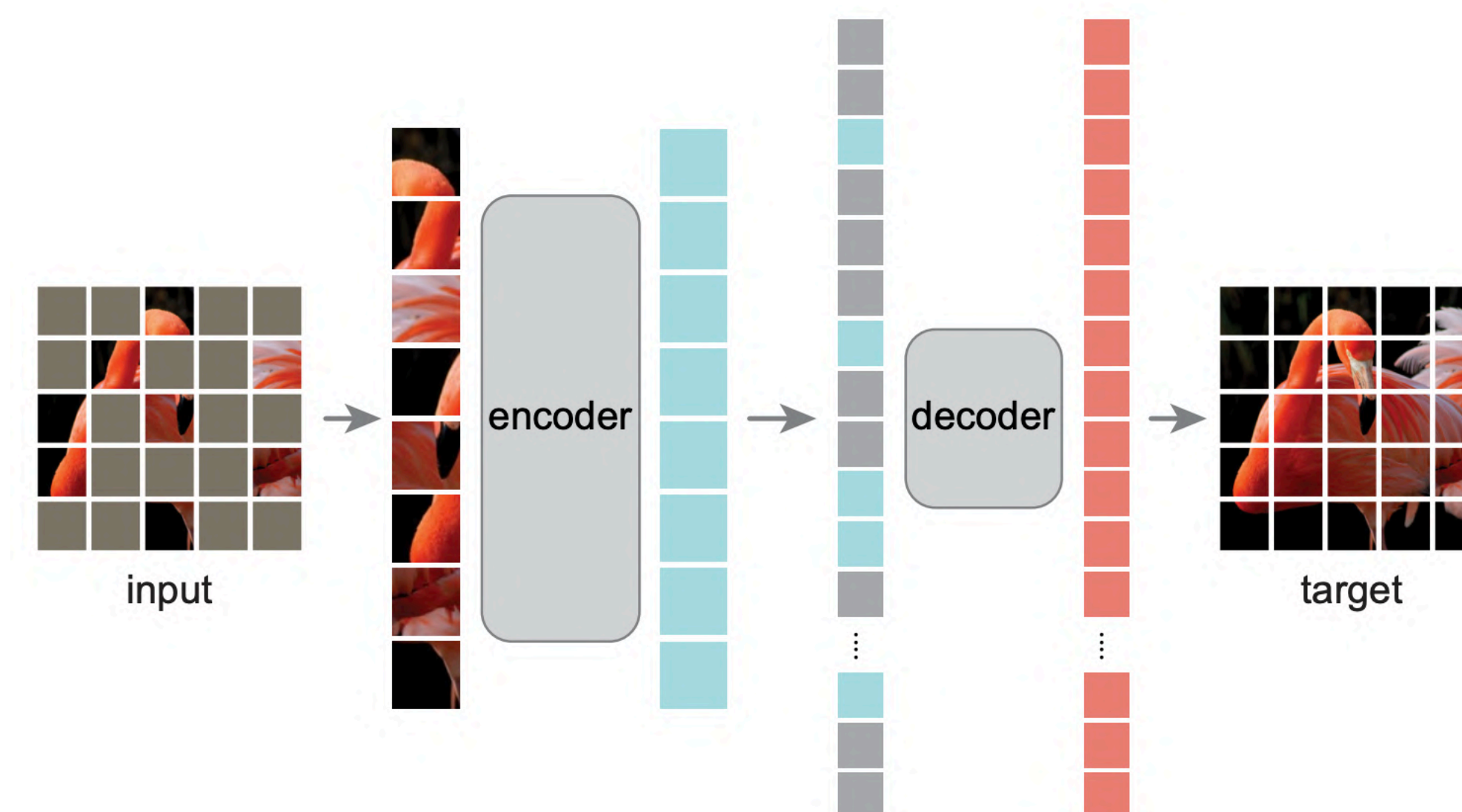
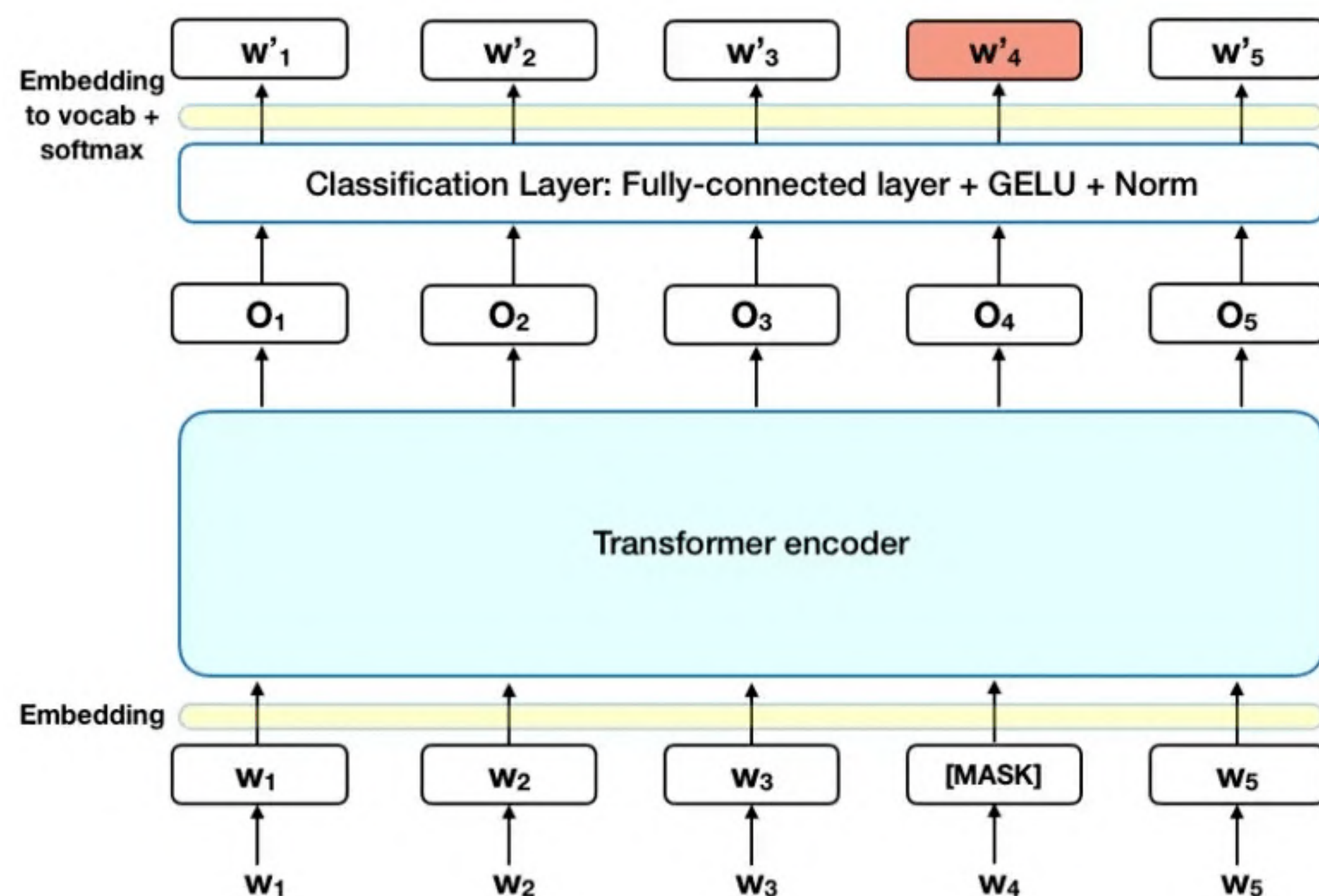


Method	Booster	ETH3D	Middlebury	NuScenes	Sintel	Sun-RGBD	Avg. Rank↓
DepthAnything (Yang et al., 2024a)	52.3	9.3	39.3	35.4	6.9	85.0	5.7
DepthAnything v2 (Yang et al., 2024b)	59.5	36.3	37.2	17.7	5.9	72.4	5.8
Metric3D (Yin et al., 2023)	4.7	34.2	13.6	64.4	17.3	16.9	5.8
Metric3D v2 (Hu et al., 2024)	39.4	87.7	29.9	82.6	38.3	75.6	3.7
PatchFusion (Li et al., 2024a)	22.6	51.8	49.9	20.4	14.0	53.6	5.2
UniDepth (Piccinelli et al., 2024)	27.6	25.3	31.9	83.6	16.5	95.8	4.2
ZeroDepth (Guizilini et al., 2023)	OOM	OOM	46.5	64.3	12.9	OOM	4.6
ZoeDepth (Bhat et al., 2023)	21.6	34.2	53.8	28.1	7.8	85.7	5.3
Depth Pro (Ours)	46.6	41.5	60.5	49.1	40.0	89.0	2.5

Foundational Models in 3D

- So far, we discussed multiple different tasks:
 - Structure From Motion
 - Point matching
 - Camera Calibration
 - Scene reconstruction
 - Multi-view pose estimation
 - Depth estimation
 - Monocular
 - Stereo

Foundational Models for NLP & 2D CV

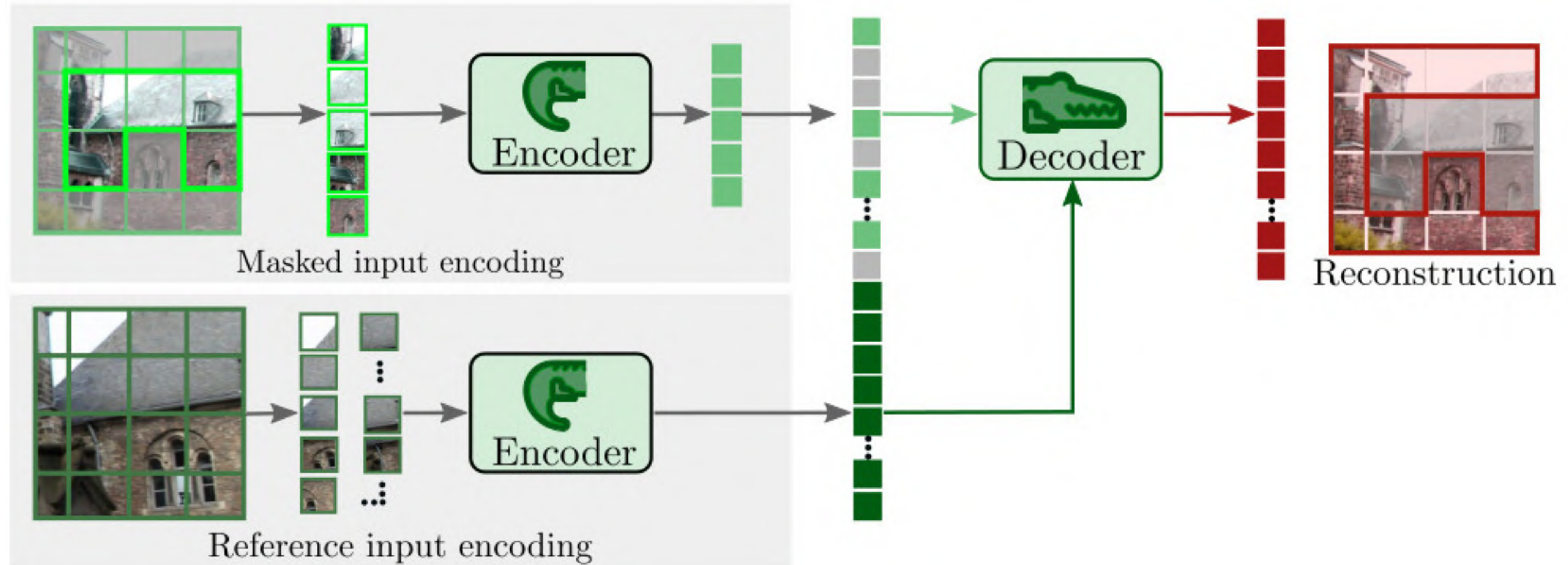


BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding <https://arxiv.org/abs/1810.04805>

Masked Autoencoders Are Scalable Vision Learners, <https://arxiv.org/abs/2111.06377>

CroCo: Same, but in 3D

- Training data: image pairs (without camera parameters)



How it Works

Reference Input
(select an image)



Mask Ratio
(adjust ratio)

90%

Masked Image
(drag to change point of view)



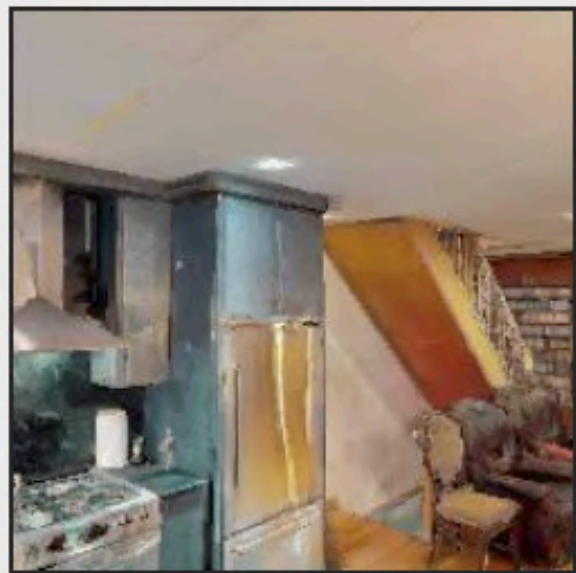
Estimated Image
(drag to change point of view)



Expected Image
(drag to change point of view)



Reference Input
(select an image)



Mask Ratio
(adjust ratio)

30%

Masked Image
(drag to change point of view)



Estimated Image
(drag to change point of view)



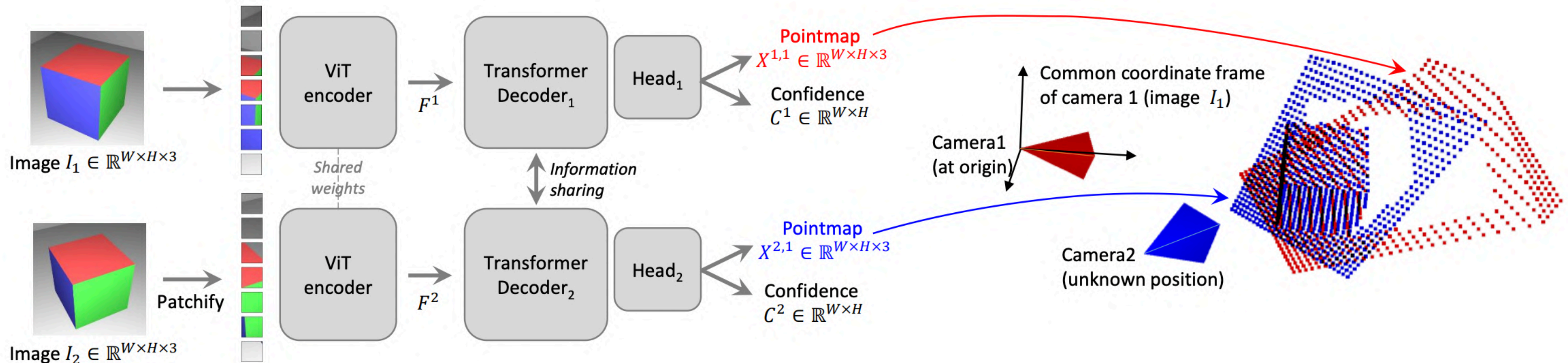
Expected Image
(drag to change point of view)

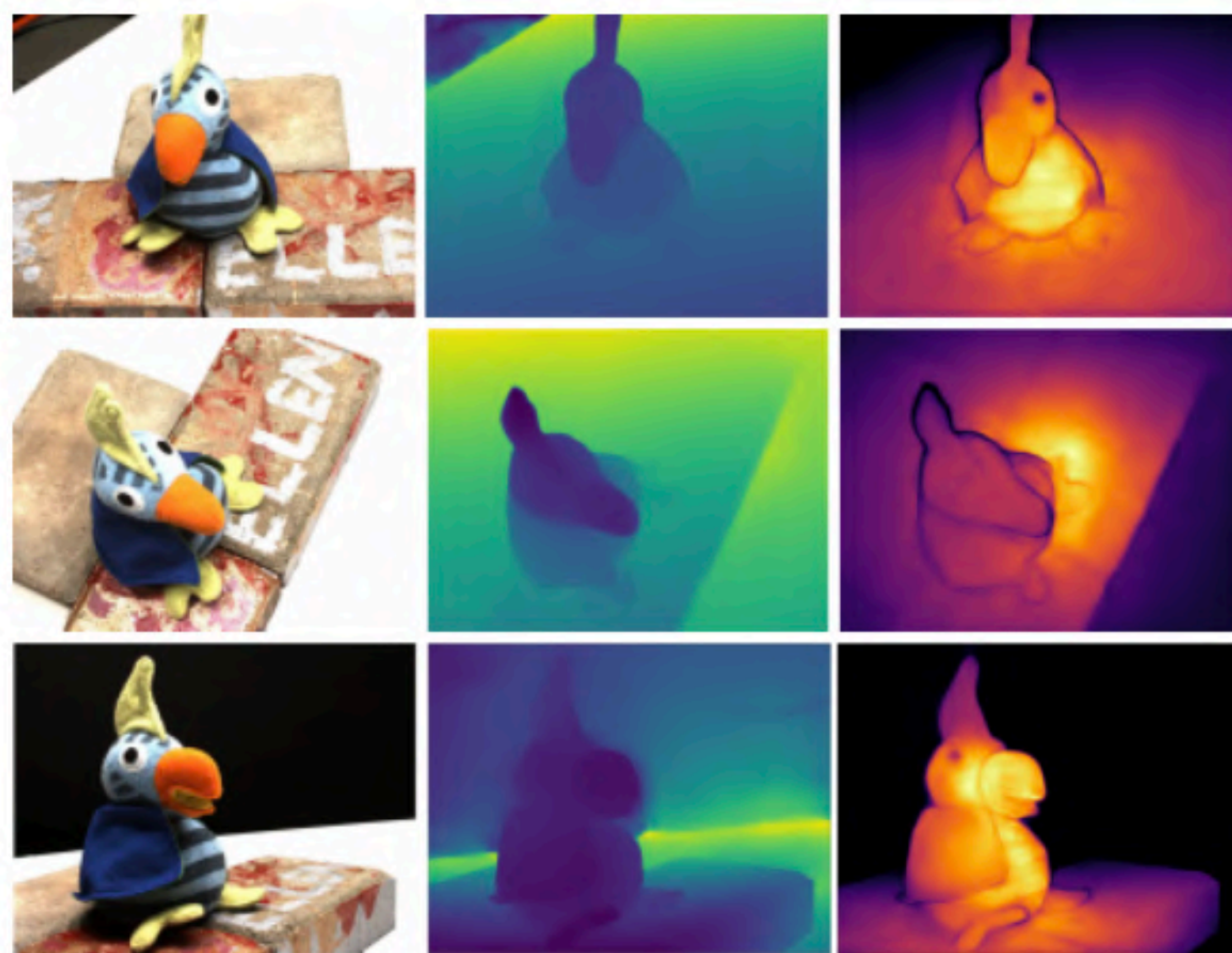
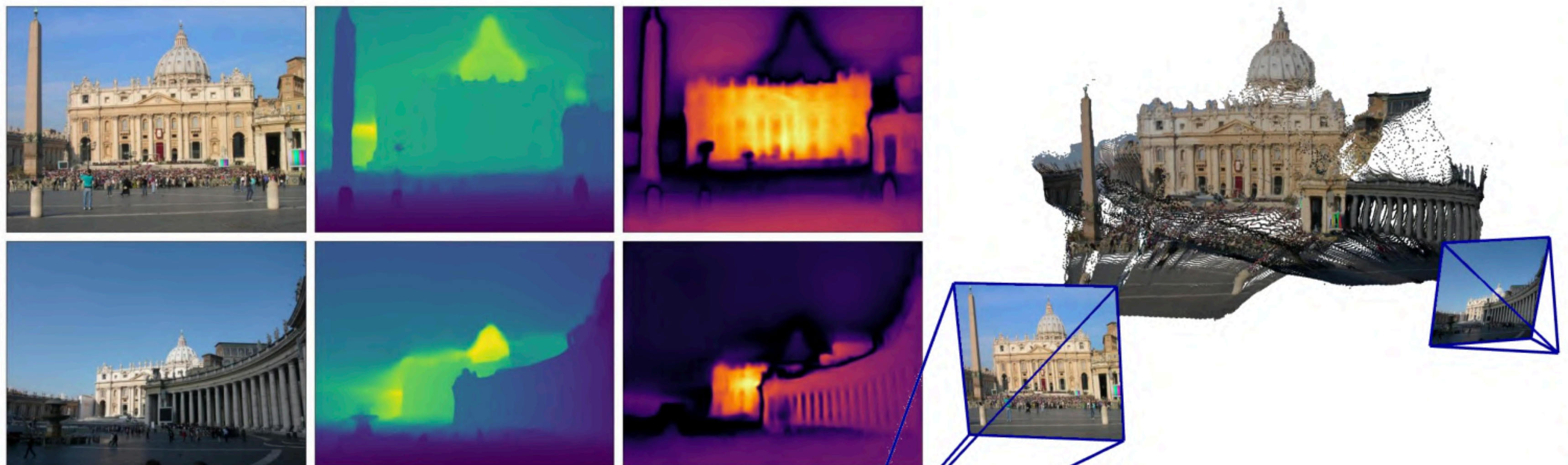


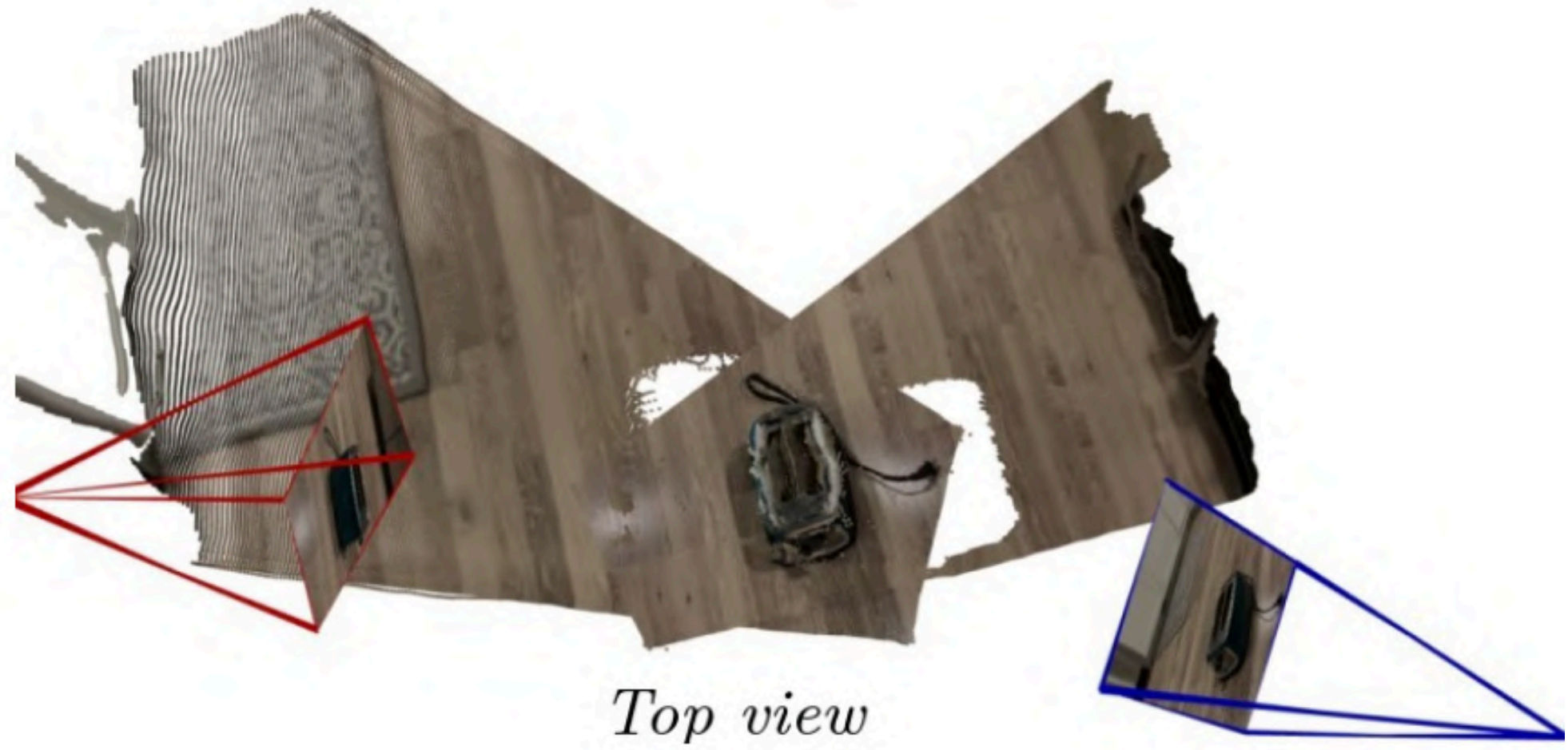
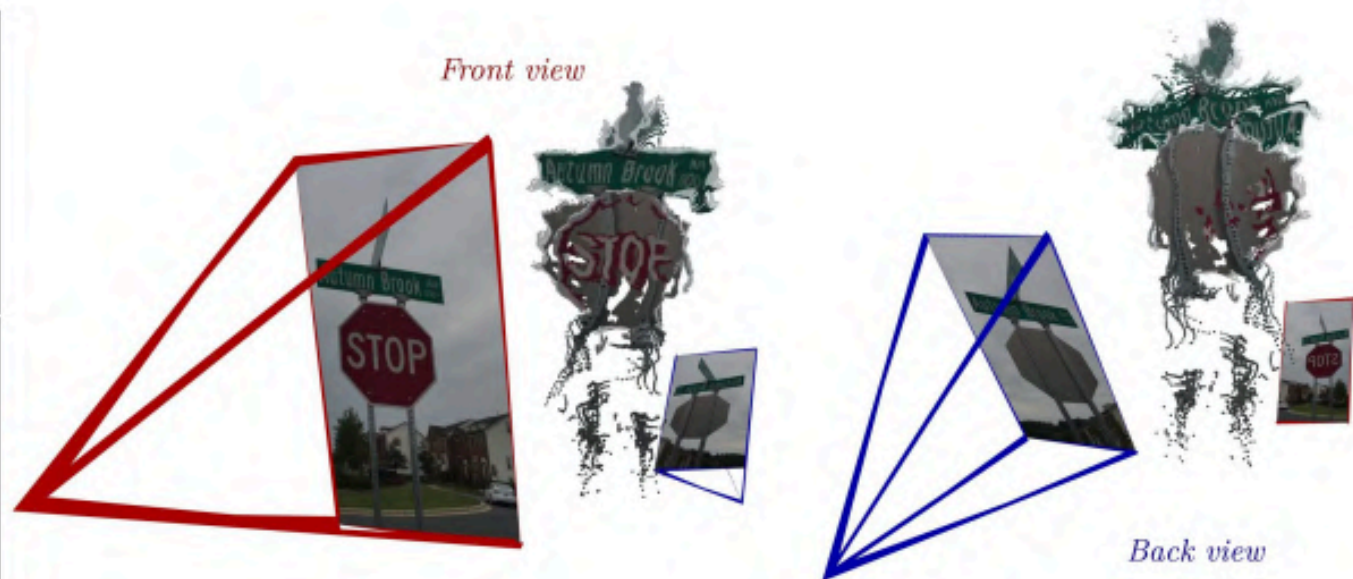
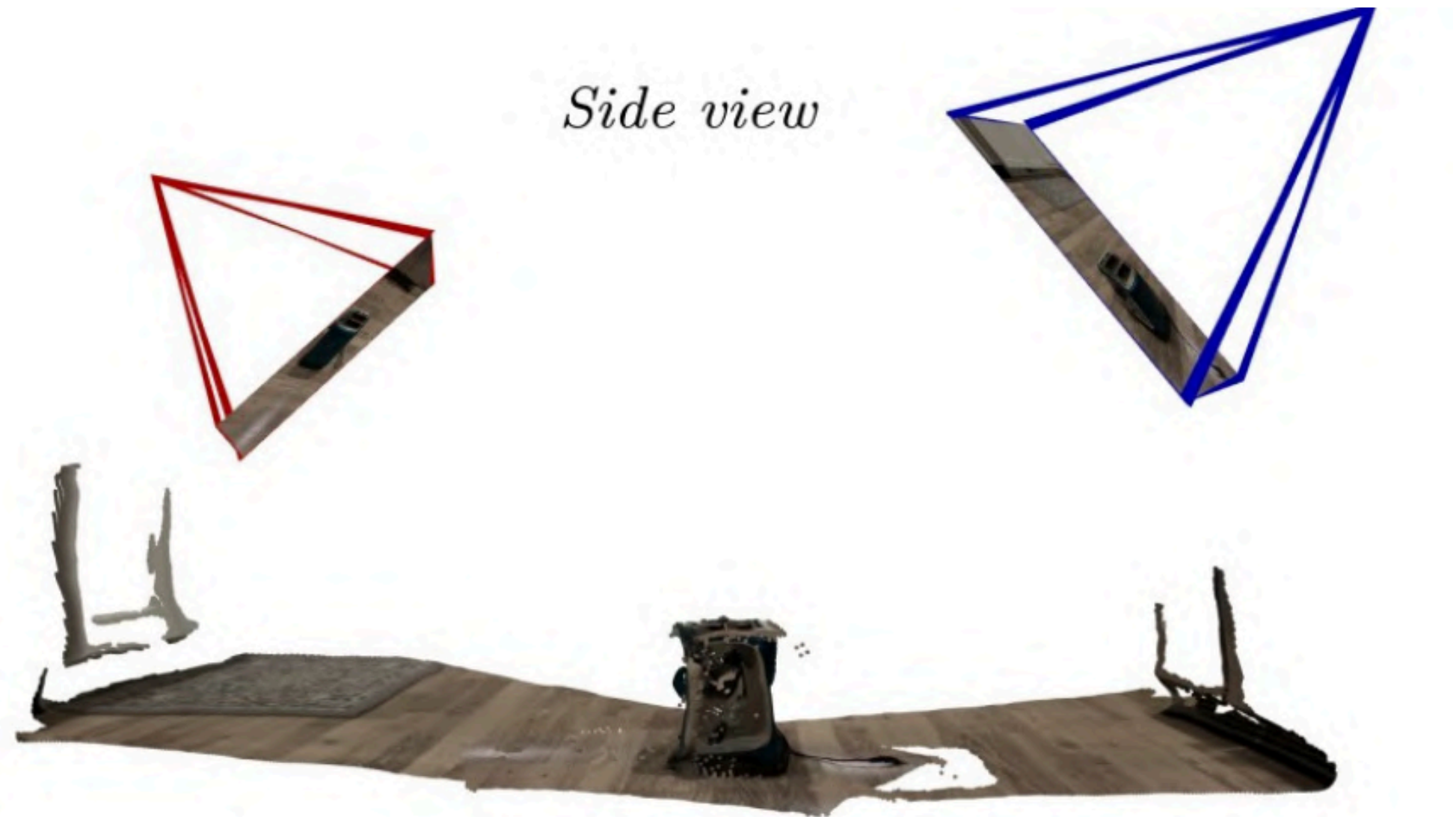
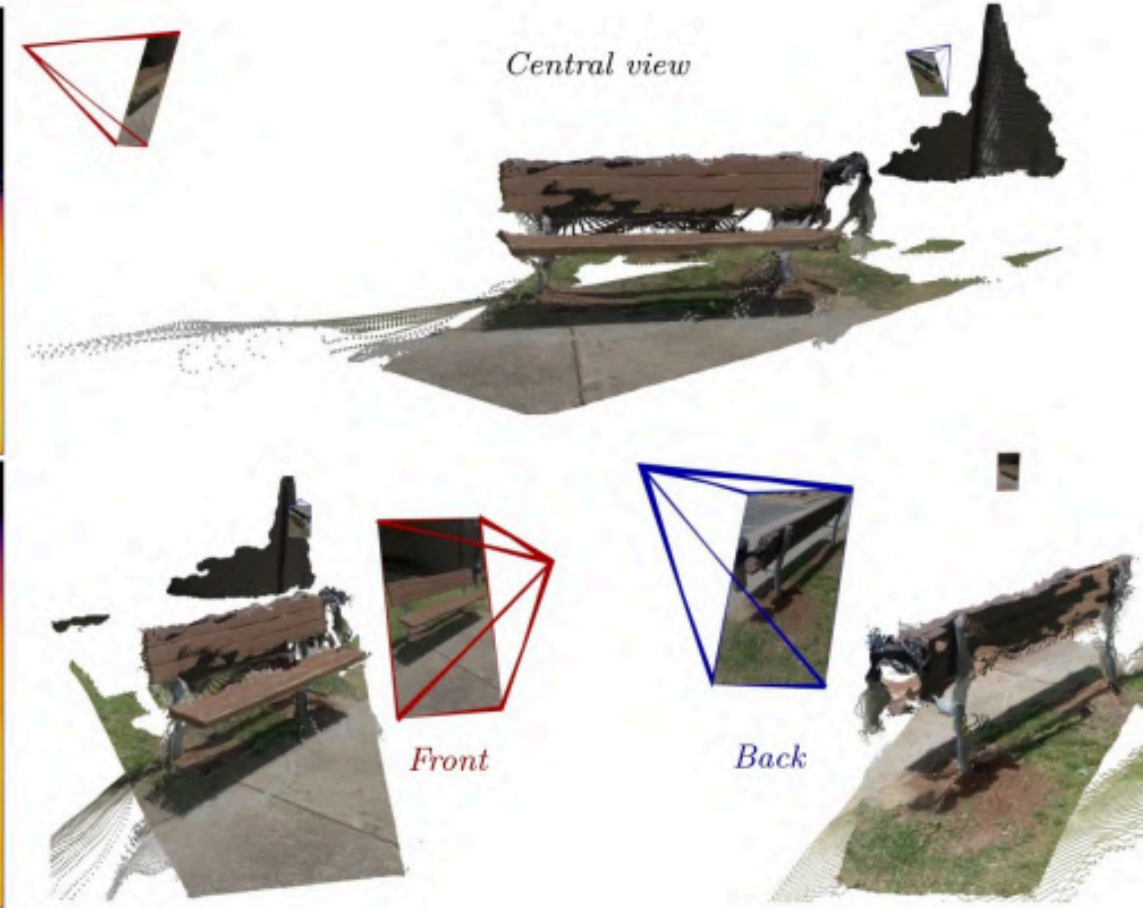
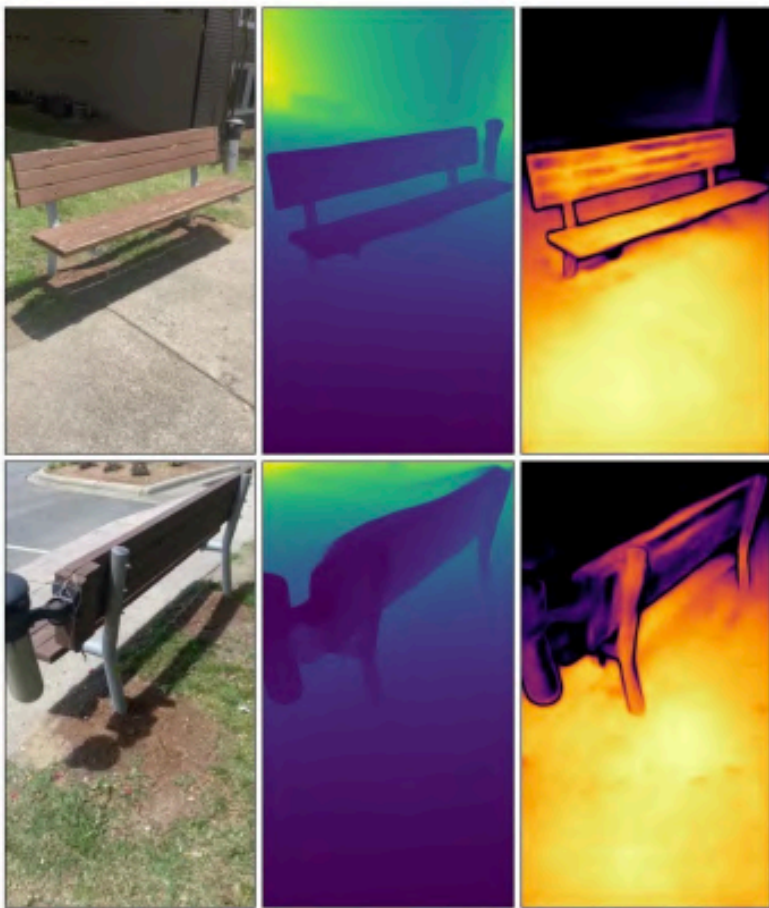
DUSt3R

Useful Tool on Top of CroCo

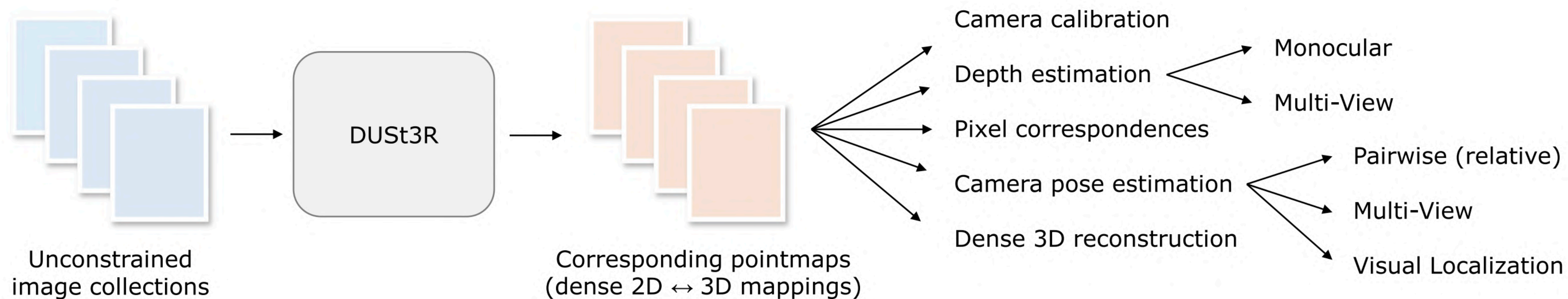
- Predict a point-map: mapping from a 2D coordinate to 3D coordinate
- Unlike CroCo, DUSt3R uses supervised objective (regression loss)







Downstream Tasks



Key Takeaways

- Depth maps allow approximating 3D structure
 - Stereo depth estimation transforms into ***disparity*** estimation
 - Scaling significantly improved monocular depth estimation
 - Foundational Models Started Appearing in 3DCV
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- Next time: recent advances in foundational models for 3D CV